

ALGORIK

GLOBAL COGNITIVE FINANCIAL AGI PLATFORM

Master Architecture & Application Blueprint

Version 11.6 - Pass-Through World Data + Native Rust Event Fabric + durable knowledge memory
Global multi-asset trading | federated world models | symbolic + neural reasoning | proactive ambient
models | native Rust event fabric | causal agency | world + tick learning | quantum-assisted research

Status: This is the target architecture. It deliberately separates capabilities that are practical today from experimental research paths. "Financial AGI" describes the product direction: a system that can observe, reason, learn, plan and operate across financial domains under hard deterministic controls; it is not a claim that present-day AGI has been achieved. Causal agency is a governed target capability: the system may optimize authorized interventions, but truthful identity/disclosure and anti-manipulation controls are architectural invariants.

1. Executive Summary

Algorik is not a trading bot and not merely an investment platform. The intended system is a global cognitive financial operating system that continuously senses the world, builds and updates a causal world model, generates hypotheses, trains specialist models, allocates capital, manages assets and liabilities, creates and trades markets, executes across regions, learns from every action and non-action, and maintains one authoritative multi-asset ledger of truth. Its intelligence is designed to be open-ended rather than fixed: the platform can discover new information domains, extend its ontology and memory, create specialist brains, acquire new tools, generate its own research curriculum, evolve models, learn from regional reflex experience and promote measured improvements through a controlled evaluation pipeline. Its learning system is explicitly dual-stream: world data teaches the platform why conditions and prices may change, while tick/order-book data teaches it how liquidity, price formation, venue behavior and market reactions actually unfold. Those streams are fused into one opportunity and decision model. The cognitive architecture is explicitly multi-model: it maintains numerous concurrent world models at different domains, regions, horizons and hypothesis branches rather than forcing all reality into one monolithic representation. A neuro-symbolic reasoning fabric connects neural perception and generative models to causal graphs, rules, temporal logic, constraints, probabilistic reasoning and executable proofs. An ambient model mesh runs continuously in the background, proactively forecasting, searching for weak signals, detecting contradictions, generating scenarios and waking specialist brains before a human or strategy asks a question. It is also an agentic causal system: beyond forecasting what may happen, it can reason about which authorized actions could change the probability of a desired outcome, simulate those interventions, execute them through governed tools, and learn their measured causal effect.

The architecture has two fundamental nervous systems. The slow cognitive system searches, reads, verifies, models, simulates, reasons and trains. The fast reflex system receives compressed knowledge, models, policy and capital envelopes, then reacts to local market state in microseconds to milliseconds without waiting for the global brain. Between them sits a warm coordination layer for cross-region arbitrage, capital movement, portfolio management, market creation and settlement. Inside those nervous systems are two complementary learning fabrics: World Intelligence and Market Microstructure Intelligence. Raw ticks are not poured indiscriminately into the largest AGI model; they are normalized, reconstructed, replayed and converted into learned microstructure representations that can be fused with world beliefs and distilled into reflex models. A third connective capability - the Causal Agency & Intervention Engine - turns beliefs and goals into bounded action plans across communications, capital, products, operations, research and permitted market actions, then feeds observed effects back into the world-model federation.

The major correction to v10.1 is architectural rather than cosmetic: intelligence is no longer "global and runs once," regional nodes are no longer isolated except for policy and outcomes, and quantum is no longer restricted to allocation. Instead, intelligence is a federation: a global cognitive core plus specialist brains plus regional reflex intelligence. Regional nodes communicate through a low-latency Reflex Mesh, enabling coordinated 2-20-leg opportunities while preserving local execution autonomy. Quantum is a hybrid research/optimization accelerator used where measurable advantage exists, including portfolio and path optimization, graph problems, feature/ensemble selection, selected quantum machine-learning experiments and simulation research. v11.6 further makes cognition polycentric: world-model populations may disagree and branch, symbolic reasoners preserve explicit invariants and proof traces, and proactive ambient models continuously maintain forecasts and surprise scores. Quantum resources can assist selected combinatorial search, scenario/ensemble selection and experimental reasoning workloads, but never confer truth or bypass evidence and deterministic controls.

1.1 Product Identity

- Global: operates across regions, time zones, currencies, jurisdictions, venues and information ecosystems.
- Multi-asset: supports liquid, event-driven, illiquid, private, tokenized and physical assets through class-specific adapters and lifecycle engines.

- Cognitive: maintains world, causal, episodic, semantic and self models; explicitly represents uncertainty and contradiction.
- Autonomous: discovers data, generates hypotheses, runs research, trains models, promotes strategies, deploys artifacts, allocates capital and operates the platform inside bounded authority.
- Reflexive: regional nodes make fast decisions locally and share compressed opportunity/state messages with peers.
- Quantum-assisted: quantum participates in selected optimization and learning workflows, always benchmarked against classical baselines.
- Financial operating system: asset management, capital management, risk management, treasury/custody, accounting/ledger, execution, settlement, market making, market creation and commerce are first-class capabilities.
- Open-ended intelligence: no fixed source list, ontology, specialist roster, model family or tool catalog is treated as permanent; gaps can generate new research, tools, agents, models and knowledge structures through governed promotion.
- Dual-stream learner: world/causal intelligence learns why markets may move; tick/order-book intelligence learns how prices, liquidity and execution react. The two are joined through time-aligned state, embeddings, beliefs and outcome labels.
- Multi-world cognition: maintains a federation of domain, regional, horizon, scenario and counterfactual world models; disagreement is represented explicitly instead of collapsed prematurely.
- Neuro-symbolic reasoning: neural models propose perceptions, hypotheses and representations while symbolic logic, constraints, causal structures and proof traces verify, combine or reject them.
- Causal agency: represents desired outcomes explicitly, identifies controllable causal levers, generates and simulates interventions, executes only through authorized action surfaces, and learns which actions actually changed outcomes versus what would have happened anyway.
- Ambient proactive intelligence: always-on models continuously forecast, simulate, monitor assumptions, search for weak signals and opportunities, and activate specialists without waiting for a prompt.
- Native Rust nervous system: internal publish/subscribe, durable event journals, control propagation and replay use the Algorik Rust Event & Control Fabric; Google Pub/Sub and Kafka are not core internal dependencies.

1.2 Non-Negotiable Safety/Correctness Boundary

1.3 External World Data Is Pass-Through; Knowledge Is Durable

Algorik must not become a warehouse of the world's raw external content. Web pages, filings, news, social/public discussion, product pages, regulatory records, geospatial feeds and other third-party world sources are viewed and processed as transient inputs. The system extracts evidence and knowledge, records how to find the source again, then discards the external content.

- Persist the source reference, provider/source identity, retrieval timestamp, version/etag when available, cryptographic hash/fingerprint, entitlement, geography, parser/version lineage and freshness/expiry metadata.
- Persist compact knowledge: normalized facts, entities, relationships, events, causal edges, embeddings/features, summaries, beliefs, confidence, contradictions, novelty, timestamps, world-model deltas and memory episodes.
- Historical reasoning re-fetches the original source or licensed historical API using its manifest. If the original content cannot be retrieved, Algorik still retains the derived knowledge plus the fact that the evidence is no longer re-fetchable.

- High-rate tick/order-book data and Algorik's own financial/execution records are separate retention classes. They may be retained when venue rights, legal obligations and policy allow because replay, accounting, audit and execution research require them.
- Ambient source watchers maintain cursors, last-seen fingerprints and knowledge deltas so Algorik can continuously know what is changing without keeping the source corpus.
- The durable knowledge layer is intentionally cheaper and smaller than source data: structured graph state, compact protobuf/Parquet knowledge packs, vector/features, summaries and episodic memory rather than copied documents or web archives.

AI and AGI can originate hypotheses, models, strategies, forecasts, explanations, opportunity intents and capital recommendations. They may also act automatically inside a pre-authorized operating envelope. However, money movement, leverage, exposure, settlement and venue access remain bounded by deterministic, testable controls. This preserves the ability to make the intelligence more general without making the blast radius general. Market-data training must preserve event time, sequence, venue identity and entitlement lineage and must prevent look-ahead leakage, timestamp contamination and impossible fills in replay. Ambient models may proactively originate research, warnings, opportunity candidates, simulations and proposed actions, but their continuous operation does not increase their authority: symbolic verification, evidence policy, risk envelopes and deterministic money-moving controls still define what can execute. External intervention is held to the same standard: truthfulness, provenance, identity, disclosure, market-conduct rules, channel permissions and jurisdiction are checked before action. The platform does not use fabricated identities, fake consensus, deceptive content, undisclosed promotion, wash activity, spoofing, rumor engineering or other manipulative conduct to manufacture an outcome.

2. Corrections Required to v10.1

v10.1 rule/limitation	v11.6 correction
Central intelligence runs once	Replace with federated cognition: global brain + specialist brains + regional reflex intelligence. Global state is authoritative for long-horizon knowledge; regions retain local state and can peer-share fast state.
Policy down, outcomes up only	Too restrictive. Keep signed policy/artifacts downward, but add peer reflex messaging, distributed reservations, opportunity tokens and cross-region outcome/state exchange.
Quantum only optimizes allocation/path decisions	Expand to a Quantum Foundry for optimization, graph workloads, feature/ensemble selection, quantum kernels/variational experiments, quantum reservoir/extreme-learning research and scenario exploration.
Rust-only machine learning	Change to Rust-first execution, not Rust-only research. Use Python/JAX/PyTorch where appropriate for research/training; export hardened inference artifacts to Rust reflex nodes.
Small models only	Keep tiny reflex models for hot paths, but allow large multimodal/foundation/generalist models in the slow lane, with distillation into fast models.

v10.1 rule/limitation	v11.6 correction
Regional nodes are independent	Replace with autonomous but connected reflex cells. No synchronous global dependency on the hot path, but nodes communicate via peer mesh for cross-region opportunities.
Physical assets are just another class	Add a Commerce & Physical Operations plane for product discovery, purchasing, logistics, inventory, resale, returns, taxes/duties and marketplace settlement.
Capital engine is a component	Promote Asset Brain, Capital Brain, Risk Brain, Treasury Bank, Ledger Brain and Settlement Brain into first-class autonomous domains with explicit contracts.
Data discovery finds sources	Upgrade to a global Scout Fabric that searches continuously, scores novelty, validates access rights, creates adapters, measures source alpha and retires weak feeds.
Data confidence is enough	Add Evidence/Truth Fabric: provenance graph, corroboration, contradiction resolution, cryptographic/source authenticity, time/location consistency and adversarial-source detection.
Implicit/fixed intelligence expansion	Add an Intelligence Expansion Engine: novelty/gap detection, self-generated curriculum, source/ontology/tool/brain registries, autonomous research campaigns, model evolution, reflex learning, collective intelligence and benchmark-gated capability promotion.
Market ticks treated mainly as execution input, not a full learning corpus	Add a Market Intelligence & Tick Learning Fabric: immutable tick/order-book history, clock normalization, book reconstruction, cross-venue alignment, replay/simulation, microstructure training, world+market fusion and reflex-model distillation.
Single world/causal model is the primary cognitive representation	Replace with a World Model Federation: many concurrent domain, regional, horizon, scenario and counterfactual models plus an arbitration layer that preserves disagreement, confidence and lineage.
Reasoning is primarily implicit inside learned/generative models	Add a Symbolic & Neuro-Symbolic Reasoning Fabric: ontologies, causal graphs, temporal/event logic, probabilistic logic, rules, constraint solvers, planning and proof traces coupled to neural models.
Models are mainly invoked by research/trading workflows	Add a Proactive Ambient Model Mesh: always-running models maintain forecasts, surprise scores, anomaly/opportunity scans, scenario branches and assumption health; they wake specialist brains automatically.
Quantum is mainly optimization + experimental QML	Expand quantum research to selected symbolic/combinatorial search, world-model/scenario ensemble selection and hypothesis-space exploration, always benchmarked against classical solvers and never treated as a truth oracle.

v10.1 rule/limitation	v11.6 correction
Prediction and planning do not explicitly model how Algorik can change outcomes	Add a Causal Agency & Intervention Engine: explicit goals, action-affordance graphs, structural causal intervention models, policy-constrained planning, authorized action tools, counterfactual no-action baselines and post-action causal attribution. The system learns not only what happens, but which lawful actions reliably change what happens.

3. Full Capability Map

Capability	Required scope
Global sensing	Web/API/WebSocket/file/stream/database/registry/satellite/geospatial/IoT/shipping/customs/blockchain/news/social/product/filing discovery and ingestion.
Evidence & trust	Source identity, provenance, licensing, freshness, cross-source corroboration, contradictions, manipulation detection, confidence and audit trail.
World model	Temporal entity/event/relationship graph spanning companies, people, assets, supply chains, countries, products, venues, contracts, weather, logistics and markets.
Financial AGI core	Planning, research, tool use, hypothesis generation, causal reasoning, counterfactuals, self-model, uncertainty, memory and specialist-agent orchestration.
Model foundry	Foundation/specialist models, supervised/unsupervised/RL, distillation, ensembles, regime models, microstructure models, valuation, risk and reflex models.
Quantum foundry	Hybrid QAOA/QUBO optimization, graph/search experiments, feature/ensemble selection, kernels/variational models, quantum reservoir/extreme-learning research, benchmark harness.
Strategy factory	Generate, backtest, simulate, stress, paper trade, promote, decay and retire thousands of strategies.
Asset management	Portfolio construction, mandates, horizons, valuation, lifecycle, corporate actions, commitments, divestment, hedging and attribution.
Capital management	Funding, liquidity ladder, collateral, margin, leverage, cash buffers, capital grants, cross-region placement, compounding and idle-cash optimization.
Risk management	Pre-trade, post-trade, market, liquidity, counterparty, concentration, causal-driver, settlement, model, data, operational and systemic risk.

Capability	Required scope
Treasury/capital bank	Cash and collateral fabric, custody corridors, FX conversion, internal funding, financing/credit interfaces where licensed, yield routing and treasury rebalancing.
Ledger/accounting	Event-sourced double-entry multi-asset ledger, positions, cash, obligations, inventory, tax lots, fees, accruals, margin, commitments and settlement state.
Reflex execution	Regional low-latency nodes with local market data, order books, features, models, strategies, risk, netting, inventory and venue adapters.
Reflex mesh	Peer-to-peer regional state/opportunity exchange, synchronized clocks, distributed reservations, opportunity epochs and bounded-loss multi-leg sagas.
Arbitrage	2-20 leg, cross-venue, cross-region, cross-asset, cross-currency, spot/derivative/tokenized/physical/event representations with size-aware graph costs.
Market making	Continuous quoting, inventory skew, adverse-selection models, rate control, dynamic spread and portfolio-aware quoting.
Market creation	Create/list/seed markets or instruments only where venue and jurisdiction permit; bootstrap liquidity and continuously risk-manage the market.
Prediction/event markets	Research, valuation, trading, market making and market creation for event contracts; regulated wagering adapters isolated by jurisdiction/eligibility policy.
Physical commerce	Discover products, buy, route, hold inventory, ship, insure, handle duties/taxes, list/resell, settle and learn SKU-level economics.
Operations/AIOps	Self-observing platform, SLOs, anomaly detection, incident triage, auto-remediation, deployment verification and autonomous test loops.
Intelligence expansion engine	Continuously detect knowledge/capability gaps; expand sources, ontology, memory, specialist brains, tools, models and research curricula; benchmark and promote improvements without requiring a redesign of the platform.
Collective intelligence fabric	Global cognition, specialist brains, evaluation agents and regional reflex cells exchange compressed knowledge, episodes, residuals, model lessons and opportunity outcomes through typed contracts and lineage.
Market intelligence & tick learning	Tick/trade/quote/L1/L2/L3/auction/options/futures/FX/crypto/funding/liquidation/on-chain and Algorik execution data; canonical event-time normalization, order-book reconstruction, microstructure features, replay, training and reflex distillation.
Historical replay & market simulation	Deterministic event-time replay with venue microstructure, latency/queue/fill assumptions, fees, rejects, partial fills, market impact and multi-leg saga simulation; supports counterfactual execution and synthetic rare-regime generation.

Capability	Required scope
World + market intelligence fusion	Time-align causal/world beliefs with market state and learned tick embeddings so the system learns both why an event matters and how that information propagates through assets, venues and liquidity.
World Model Federation	Concurrent global, macro, geopolitical, company, supply-chain, physical, microstructure, event, portfolio/capital, counterparty, regional, self, scenario and counterfactual models; models can branch, disagree, merge, expire and be scored by predictive calibration.
Symbolic & neuro-symbolic reasoning fabric	Knowledge graph + ontology reasoning, temporal/event calculus, causal reasoning, rules, probabilistic logic, constraint/SAT/SMT/MILP/graph solvers, abductive reasoning, planning, proof traces and neural-to-symbolic compilation.
Ambient proactive model mesh	Always-on background models continuously predict, detect novelty, monitor assumptions, simulate future paths, scan weak signals, identify opportunity/risk precursors and route attention to the correct specialist or reflex lane without waiting for prompts.
World-model arbitration & epistemic control	Compare competing model states/hypotheses, preserve minority explanations, score calibration/evidence fit, detect contradictions, maintain uncertainty and decide when to branch, merge, abstain or request more information.
Causal agency & intervention engine	Desired-outcome models, controllable-variable/affordance graph, do-calculus/structural causal intervention reasoning, action generation, multi-step planning, simulation, policy/authority gates, execution orchestration and closed-loop causal attribution.
Authorized action surfaces	Truthful publishing/communications, authorized channel posting, stakeholder outreach, research/data acquisition, product/pricing/configuration changes, operational actions, capital/hedging/liquidity actions, commerce actions and permitted market/product creation - each exposed as a typed tool with identity, scope, disclosure and audit policy.
Intervention learning & effect attribution	Estimate treatment/uplift/causal effect over multiple horizons using experiments where appropriate, observational causal inference where experiments are impossible, synthetic controls, counterfactual world-model branches and no-action baselines; update causal graphs and action policies from measured effect rather than correlation alone.

4. Time-Lane Architecture

The platform should be organized by time budget. A function belongs in the fastest lane that can meet its correctness requirements, not the fastest lane that sounds impressive.

Lane	Time scale	Responsibilities
Lane 0 - Reflex Hot	~10 us to 10 ms	Local feed decode, book update, features, reflex inference, strategy predicates, local risk, netting, routing, order/quote generation. No remote dependency. Uses compact tick-derived features/embeddings and distilled microstructure/reflex models; never reads the historical Tick Lake in the order path.
Lane 1 - Coordinated Fast	~1 ms to seconds	Peer reflex messages, multi-region opportunity tokens, distributed inventory reservations, multi-leg coordination, hedge/unwind commands. Cross-region latency is accepted explicitly.
Lane 2 - Tactical/Warm	seconds to minutes	Portfolio/risk refresh, capital placement, treasury moves, market-making regime shifts, event repricing, cross-region inventory rebalancing. Aggregated microstructure state, cross-venue liquidity maps, execution-quality refresh and short-horizon world/market fusion. Includes bounded tactical interventions such as authorized pricing/configuration, inventory/capital rebalancing and pre-approved operational actions.

Lane	Time scale	Responsibilities
Lane 3 - Cognitive Slow	minutes to days	Global scraping/discovery, evidence synthesis, causal reasoning, research, hypothesis generation, simulation, strategy discovery, model training and validation. Historical tick replay, order-book reconstruction, microstructure training, representation learning and world+market fusion. Hosts long-running ambient world models, symbolic reasoners, competing hypothesis branches, proactive forecasting and model-arbitration cycles. Also performs causal intervention planning, communications/outreach planning, action simulation, experiment design and post-action effect attribution.
Lane 4 - Quantum/Research	minutes to days	Quantum-classical optimization and learning experiments, benchmarked asynchronous workloads, scenario search and novel algorithm research. Never blocks execution. May benchmark feature/ensemble/path-selection or hybrid learning ideas against classical microstructure baselines; never blocks execution. Includes experimental quantum-assisted symbolic/combinatorial search, scenario diversification and world-model ensemble selection where benchmarked value is demonstrated. May explore large combinatorial intervention sets or scenario branches asynchronously; quantum output is advisory and classically verified.

5. End-to-End System Path

WORLD / ECONOMIC / GEOPOLITICAL / COMPANY / PHYSICAL DATA

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GLOBAL SCOUT -> EVIDENCE & TRUTH -> WORLD MODEL / CAUSAL MEMORY

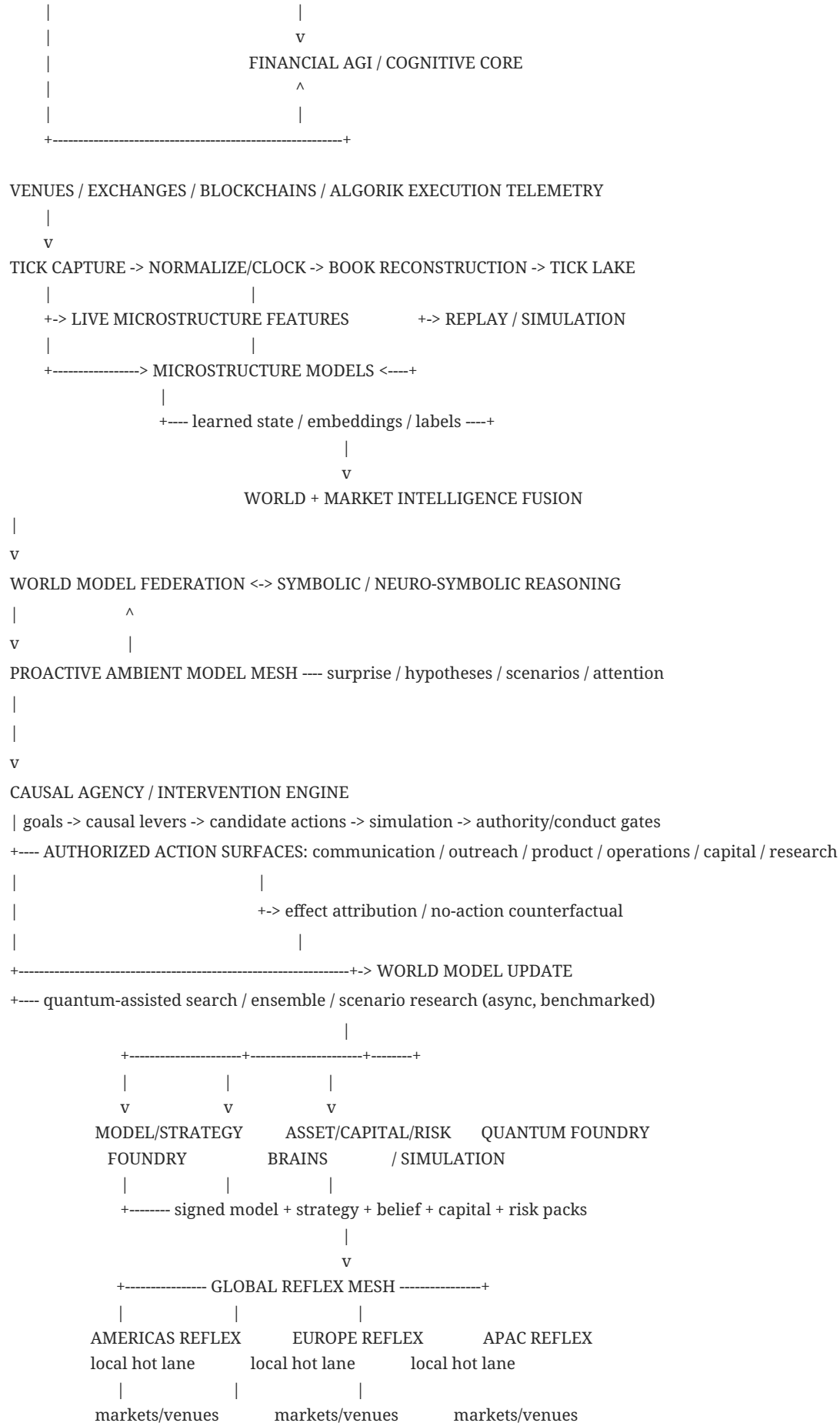
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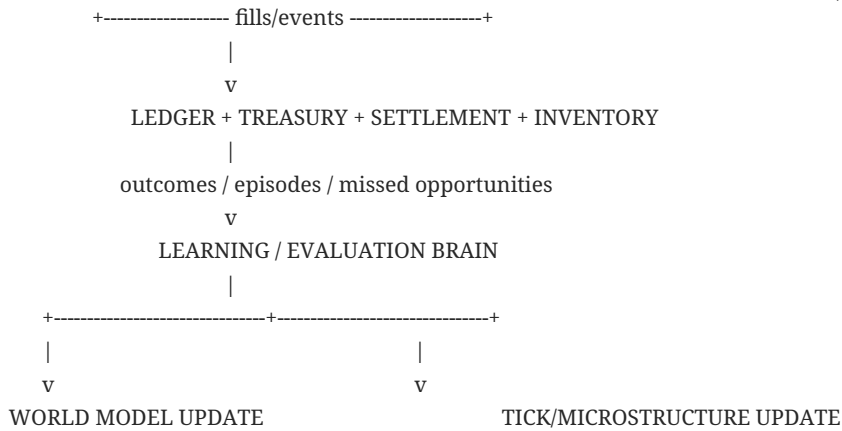
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Target architecture - global multi-asset cognitive trading, capital and market system





6. Global Scout & Data Acquisition Fabric

The Scout Fabric is a continuously operating discovery system. It does not wait for a human to add a feed. It searches for new lawful sources, measures whether they add unique predictive information, creates or proposes adapters, monitors access/terms, and retires low-value sources. The goal is not to hoard the internet; it is to turn the changing world into structured evidence with provenance.

6.1 Source Universe

- Exchange feeds, order books, trades, quotes, options surfaces, funding, borrow, rates and venue status. Raw market events feed both the live reflex path and the historical Market Intelligence & Tick Learning Fabric, subject to venue entitlements and retention rules.
- Filings, earnings, regulatory portals, court/docket data, procurement, patents, corporate registries, government statistics and central-bank releases.
- News, press releases, expert publications, public web, deep-web databases, public social signals and public discussion communities.
- Blockchain state, mempools, token flows, smart-contract events and protocol governance.
- Geospatial/satellite, weather, shipping/AIS, flight, port, traffic, commodity flows, energy grids and public IoT where licensed/available.
- Product/SKU listings, inventories, prices, shipping estimates, marketplace fees, auction data and resale markets.
- Private/alternative/licensed datasets and user-authorized sources under explicit entitlements.

6.2 Discovery Agent Loop

6.3 Pass-Through Acquisition and Knowledge Distillation

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SOURCE / API / WEB / FEED -> TRANSIENT FETCH -> PARSE / EXTRACT -> EVIDENCE CHECK -> ENTITY / EVENT / FACT / FEATURE
-> KNOWLEDGE DELTA -> WORLD MODELS / MEMORY
      \-> SOURCE MANIFEST + HASH + CURSOR
      \-> RAW EXTERNAL CONTENT DISCARDED
  
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The Scout Fabric is a live sensory system, not a data hoarder. External source payloads exist only long enough to parse, validate, extract and reason over them. Durable storage begins at the knowledge boundary, not the document boundary.

1. Observe gaps in the world model, strategies and forecast errors.
2. Generate source-discovery queries and candidate locations by geography, entity, domain and language.
3. Inspect access method, law/terms, schema, freshness, cost, expected uniqueness and manipulation risk.

4. Register approved source metadata and instantiate a feed/adaptor in sandbox.
5. Score the source against incremental information gain, timeliness, reliability and realized strategy contribution.
6. Promote, throttle, quarantine or retire the source automatically under policy.

7. Evidence, Provenance & Truth Fabric

No single source is "truth." Every extracted claim becomes an Evidence object with source identity, timestamp, location, extraction method, authenticity signals, license, confidence and links to corroborating or contradicting evidence. The system builds a provenance graph and updates beliefs from evidence rather than converting scraped text directly into facts.

Mechanism	Purpose
Authenticity	TLS/domain identity, signed releases where available, document hashes, chain proofs, known issuer identity, source history.
Corroboration	Independent-source agreement weighted by common-source dependence so ten copies of one press release do not count as ten sources.
Temporal consistency	Does the claim fit prior state, event ordering and physically possible timelines?
Geospatial consistency	Location, routing, shipping, weather and jurisdiction evidence must agree where relevant.
Contradiction engine	Maintains competing hypotheses instead of deleting conflicts; tracks which later evidence resolves them.
Adversarial detection	Spam, coordinated manipulation, synthetic content, poisoned feeds, spoofing and anomalous source behavior.
Quantum role	Optional experimental optimization/clustering/model-selection work on evidence graphs; quantum does not itself confer authenticity or truth.

8. Market Intelligence & Tick Learning Fabric

World data teaches Algorik why markets may move. Tick and order-book data teaches it how prices, liquidity, queues, spreads, venue behavior and cross-asset reactions actually evolve. Both are required. The tick fabric is a first-class learning system, not merely a feed handler for execution.

The design deliberately separates raw market-event storage from the Financial AGI context window. Every raw tick can remain available for exact replay and retraining, while the cognitive layer consumes time-aligned market states, learned embeddings, microstructure regimes, causal reaction summaries and exception episodes. This preserves information without forcing a generalist model to ingest billions of low-value events directly.

8.1 Tick and Market-Event Universe

- Trades/ticks, bid/ask quotes, NBBO or venue-equivalent top-of-book, L1/L2 depth and L3/order-level data where licensed and economically justified.
- Auction/open/close/imbalance events, halts, circuit breakers, venue status, rejects, cancels and order acknowledgments.
- Options chains, implied-volatility surfaces, Greeks, futures curves, basis, borrow, funding and financing state.
- FX and rates quotes, crypto order books, perpetual funding, liquidations, mempool/on-chain market signals and decentralized venue state.
- Algorik internal telemetry: submitted orders, timestamps, routing decisions, fills, partial fills, slippage, queue estimates, cancels, rejects, hedges and risk vetoes.
- Cross-asset and cross-region synchronization metadata so reactions can be learned across instruments and venues rather than in isolation.

8.2 Tick Data Pipeline and Storage

Module	Responsibility
Venue capture adapters	Decode native feeds with source sequence numbers, exchange timestamps and receive timestamps. Preserve raw messages where rights allow.
Clock + sequence normalization	PTP/GPS-grade clock discipline where justified; preserve event time, receive time, sequence, venue and uncertainty rather than collapsing everything to one timestamp.
Canonical market-event schema	Normalize trades, quotes, depth updates, auctions, status and lifecycle events into typed records while retaining source-native payload hashes/lineage.
Immutable Tick Lake	Partitioned historical raw/canonical events with entitlements, retention, compression, checksums and reproducible manifests. Separate hot replay indexes from cold archive.
Order-book reconstruction	Rebuild venue books deterministically, detect gaps/out-of-order events, create snapshots/checkpoints and mark periods where reconstruction is unreliable.
Cross-venue/cross-asset aligner	Align asynchronous instruments by event time and information arrival, <u>preserving uncertainty and avoiding artificial simultaneity</u> .
Feature + representation engine	Generate spread, depth, imbalance, queue, flow, toxicity, volatility, liquidity, lead/lag, cross-venue and learned embedding representations.
Outcome/label builder	Create fill probability, future return, adverse selection, market impact, realized volatility, opportunity decay and multi-horizon outcome labels without future leakage.
Replay + simulation engine	Replay canonical historical markets and Algorik decisions with latency, queue, fees, rejects, partial fills, impact and multi-leg execution semantics.
Microstructure training service	Train/calibrate specialist, execution, liquidity, slippage, fill, regime and short-horizon models; benchmark by venue, asset, regime and horizon.
Reflex distillation/packager	Compress useful slow/warm intelligence into bounded features/models/lookup structures that regional reflex nodes can evaluate within their latency budgets.
Online drift + feedback	Measure live residuals, fill/slippage error, regime drift and venue changes; feed retraining and Curriculum Engine tasks rather than silently adapting money-moving logic.

8.3 Market Replay, Counterfactual Execution and Synthetic Experience

Replay is the bridge between historical market truth and safe autonomous learning. A strategy is evaluated against the market that existed, not against bars created after the fact. The simulator must model what Algorik could have known and done at each instant, including data arrival, decision latency, queue position, rate limits, fees, rejects, partial fills, impact and the possibility that a multi-leg opportunity disappears before completion.

- Exact historical replay: deterministic event sequencing from venue-native/canonical records with book checkpoints and integrity flags.
- Counterfactual replay: compare actions taken, vetoed or missed against feasible alternative actions using the same information set available at that time.
- Multi-leg arbitrage replay: model 2-20-leg opportunity discovery, reservation, legging risk, path decay, partial execution, hedge/unwind and capital consumption across regions.
- Agent/market simulation: generate synthetic rare regimes, liquidity droughts, manipulation patterns, venue outages and stress events to supplement sparse real history.
- Execution digital twin: continuously calibrate fill, queue, impact and latency models to Algorik live telemetry so paper/replay assumptions do not drift away from reality.

8.4 World + Market Intelligence Fusion

The two learning streams meet through a time-aligned fusion layer. World Intelligence supplies events, entities, causal hypotheses, beliefs, uncertainty and expected propagation paths. Market Intelligence supplies observed reaction, microstructure regime, liquidity, flow, lead/lag structure, pricing dislocation and execution feasibility. The fused representation lets Algorik learn not only that an event occurred, but how information is being transmitted through markets and whether a tradable reaction still exists.

- World event -> expected impacted entities/assets/venues -> watch graph and hypothesis priors.
- Tick reaction -> measured propagation, liquidity response, cross-asset timing and opportunity decay -> belief update.
- Mismatch between world belief and market behavior -> investigation, arbitrage/relative-value hypothesis or evidence-quality challenge.
- Fused state -> strategy/model foundry, Asset/Capital/Risk brains and specialist brains.
- Distilled fused state -> regional reflex models and compact opportunity maps for millisecond/microsecond decisions.

8.5 Data Correctness and Training Guardrails

- No look-ahead: training/replay may use only data that would have been observable by the decision timestamp, including realistic publication/feed arrival delays.
- No impossible fills: a replayed trade must respect available depth, queue/priority assumptions, rate limits, latency, partial fills, venue rules and price impact.
- No timestamp laundering: source time, receive time, normalized time and uncertainty remain distinct; clock corrections are lineage-tracked.
- No hidden survivorship: delisted instruments, venue outages, symbol changes, corporate actions and failed products remain represented in historical universes.
- Entitlements are part of lineage: model artifacts record which licensed market data trained them and where that artifact is permitted to run/use outputs.
- Raw tick volume is not intelligence by itself: models and the Evaluation Brain must prove incremental predictive/execution value versus simpler baselines.

9. Financial AGI / Cognitive Core

The Cognitive Core is the slow-thinking system. It is a planner and scientific engine, not the hot-path trader. It can decompose goals, launch research campaigns, call tools, coordinate specialist agents, inspect evidence, build causal hypotheses, run simulations, request new data, train models, challenge its own conclusions and produce deployable policy/model packages.

9.1 Specialist Brain Society

Brain	Scope
Macro/Geopolitical Brain	Rates, inflation, policy, elections/public policy facts, trade, conflict, sanctions, sovereign and currency regimes.
Company/Fundamental Brain	Financial statements, earnings, corporate events, management guidance, credit and capital structure.
Microstructure Brain	Books, flows, liquidity, queue dynamics, adverse selection, venue behavior and execution cost.
Event/Prediction Brain	Event probabilities, scenario trees, information arrival and event-contract valuation.
Supply Chain/Physical Brain	Products, commodities, suppliers, logistics, inventories, production constraints and shipping.
Crypto/On-chain Brain	Protocol state, token economics, bridges, liquidity, smart contracts and on-chain flows.
Private Markets Brain	Illiquid valuations, commitments, capital calls, private-company/fund lifecycle.
Portfolio/Asset Brain	Mandates, horizons, exposures, hedges, divestment, factor/causal-driver allocation.
Capital Brain	Liquidity, collateral, leverage, funding, capital grants, placement and compounding.
Risk Brain	Risk forecasting, scenario/stress generation and risk recommendations; deterministic gates enforce hard limits.
Adversarial Brain	Looks for manipulation, regime breaks, model failure, source poisoning and strategies that exploit Algorik itself.
Meta Brain	Allocates research attention, measures which brains/models are reliable in which regimes, and resolves disagreements.
World Model Arbitration Brain	Scores and reconciles competing world models/hypotheses, preserves useful disagreement, tracks calibration and decides branch/merge/abstain/request-more-data actions.
Symbolic Reasoning Brain	Runs ontology, temporal, causal, rule, probabilistic and constraint reasoning; produces explicit derivations and machine-checkable constraints.
Ambient Attention Brain	Always-on attention router that combines surprise, opportunity, risk and assumption signals and wakes the right specialist/research/reflex pathways.
Causal Agency Brain	Goal decomposition, controllable-variable discovery, action-affordance graph, intervention generation, causal simulation, multi-step planning, expected-effect/risk/cost comparison and execution orchestration through authorized tools.

Brain	Scope
Communications & Conduct Brain	Truthfulness/provenance, identity/disclosure, channel policy, audience/jurisdiction rules, market-conduct constraints and post-publication monitoring for authorized external communications. It blocks deceptive or manipulative influence patterns.

9.2 Memory & Self Model

- Semantic memory: entities, relationships, definitions, policies and stable knowledge.
- Episodic memory: what the world looked like, what Algorik believed, what it did, and what followed.
- Procedural memory: research procedures, strategy templates, data adapters, playbooks and remediation patterns.
- Counterfactual memory: what would likely have happened under rejected or alternate actions.
- Self-model: known weaknesses, blind spots, uncertainty, stale areas, model/data drift, operational health and capability boundaries.

9.3 World Model Federation

Algorik must not depend on a single monolithic world model. It maintains a population of world models specialized by domain, geography, horizon, abstraction level and hypothesis. Each model has explicit scope, evidence lineage, calibration, update cadence and expiry. Models may disagree; disagreement is information. A World Model Arbitration layer compares evidence fit, predictive performance, causal coherence and uncertainty, and can preserve multiple live branches until reality resolves them.

- Global causal model - long-horizon entities, events, drivers and cross-domain causal structure.
- Macro/geopolitical models - rates, policy, trade, conflict, sanctions, currencies and sovereign regimes.
- Company/fundamental models - firms, financials, management actions, credit and capital structure.
- Supply-chain/physical models - factories, commodities, inventories, logistics, products, weather and real-world constraints.
- Market/microstructure models - liquidity, order flow, venues, lead/lag, impact, arbitrage and execution state.
- Event/prediction models - event trees, conditional probabilities, information arrival and resolution mechanics.
- Portfolio/capital/liquidity models - exposures, collateral, funding, leverage, optionality and capital mobility.
- Counterparty/network models - brokers, exchanges, bridges, custodians, settlement dependencies and contagion paths.
- Regional world models - local language, regulation, venues, microstructure, logistics and information ecosystems.
- Scenario/counterfactual models - alternate futures, rejected actions, stress worlds and rare-regime branches.
- Self/system models - Algorik's own capabilities, failure states, blind spots, latency, data/model health and authority.

9.4 Symbolic & Neuro-Symbolic Reasoning Fabric

The system needs many forms of explicit reasoning in addition to neural inference. Neural models are excellent proposal engines and representation learners; symbolic systems are valuable for invariants, exact constraints, compositional reasoning and auditable derivations. Algorik therefore treats reasoning as a society of interoperable reasoners rather than a single chain-of-thought model.

- Knowledge-graph and ontology reasoning over typed entities, relationships, events and financial/physical lifecycles.
- Temporal and event logic for before/after, deadlines, settlement calendars, causal ordering and state transitions.
- Causal graph reasoning for interventions, propagation paths, confounders and counterfactual explanations.
- Rule and policy engines for eligibility, accounting invariants, market structure, contracts and deterministic business logic.
- Constraint reasoning using graph solvers, CSP/SAT/SMT/MILP/QUBO-style formulations where appropriate.
- Probabilistic symbolic reasoning for uncertain rules, Bayesian structures and evidence-weighted hypotheses.
- Abductive reasoning: infer the best current explanations for surprising observations while preserving alternatives.
- Planning/program synthesis: compose tools, actions, experiments and research plans from explicit goals and constraints.
- Proof/derivation artifacts: important conclusions can carry a machine-readable derivation, evidence set, violated assumptions and uncertainty rather than only free-form prose.
- Neural-to-symbolic compilation: validated discoveries can become rules, graph predicates, constraints or bounded reflex predicates that execute deterministically.

9.5 Proactive Ambient Model Mesh

Ambient models are persistent background intelligence. They do not wait for a user prompt or a strategy tick. They continuously maintain forecasts, expected-state distributions, weak-signal detectors, surprise measures and conditional scenarios across the world and markets. An Attention Router turns material deviations into research tasks, specialist activation, risk review, capital preparation or reflex-model updates.

- Forecast models continuously maintain expected future states over multiple horizons, from milliseconds to years.
- Surprise/novelty models compare live observations with expected states and surface unexplained residuals.
- Assumption sentinels continuously test whether the premises behind active strategies, beliefs and risk models still hold.
- Opportunity sentinels look for precursors to cross-asset, cross-region, event, physical-product and multi-leg dislocations before a full trade signal exists.
- Risk sentinels model tail conditions, contagion, liquidity evaporation, crowding and correlated failure before hard limits are reached.
- Counterfactual shadow models continuously simulate alternate actions and maintain learning data for decisions not taken.
- Capital-readiness models anticipate where collateral, cash, inventory or hedges may be needed and can recommend pre-positioning within authority limits.
- Curiosity models search for high-value unknowns and trigger new scouts, tools, simulations, specialists or symbolic hypotheses.
- Ambient outputs are advisory until promoted through the same evidence, evaluation, authority and deterministic-control boundaries as any other model output.

10. Model & Strategy Foundry

v11 removes the Rust-only/small-model restriction from research. The hot lane stays Rust-first and bounded; the slow lane may use the best fit research ecosystem. Large multimodal/generalist models can reason over evidence and generate hypotheses, while smaller specialist/reflex models are trained or distilled for predictable low-latency inference.

Model class	Lane	Purpose
Foundation/generalist models	Slow lane	Research, extraction, planning, tool use, explanation and hypothesis generation.
Specialist predictive models	Slow/warm	Regime, event probability, valuation, liquidity, slippage, default, volatility, demand and supply.
Reflex models	Hot	Tiny bounded inference for order-book state, fill probability, adverse selection, short-horizon signals and opportunity scoring.
Policy/distilled models	Hot/warm	Compressed outputs from larger models that can be evaluated deterministically and cheaply.
Ensembles	All non-hot or compact hot	Model diversity, calibrated uncertainty and regime-conditioned selection.
Microstructure representation models	Slow/warm	Learn compact representations of book/flow/liquidity/venue state from tick streams for fusion, regime detection and downstream specialist/reflex models.
Execution digital-twin models	Slow/warm	Fill probability, queue, slippage, impact, latency and opportunity-decay models calibrated to Algorik telemetry and historical replay.
World-model ensembles	Slow/warm	Concurrent domain/regional/horizon/scenario models with arbitration, disagreement and calibration.
Symbolic / neuro-symbolic reasoners	Slow/warm	Rules, graphs, temporal/causal logic, probabilistic logic, constraints, planning and neural-symbolic hybrids.
Ambient forecasting / surprise models	All lanes by distilled form	Persistent expected-state, anomaly, precursor, assumption and opportunity models that run continuously.
World-model/reflex compilers	Slow -> warm/reflex	Translate validated symbolic predicates, representations and model outputs into bounded deployable artifacts.
Intervention-effect / uplift models	Slow/warm	Estimate heterogeneous causal effect of candidate actions on target outcomes across entities, audiences, markets and horizons; support treatment selection and abstention when effect is not identifiable.

Model class	Lane	Purpose
Action-policy / planner models	Slow/warm	Propose sequenced interventions under causal, cost, risk, legal, identity, authority and timing constraints; evaluated against simulators and counterfactual world-model branches before authority is granted.

10.1 Training Lifecycle

1. Data campaign is assembled from retained internal episodes plus reproducible external references and licensed extracts.
2. Features/representations are computed with strict lineage and leak prevention.
3. Classical baseline is trained first and calibrated.
4. Quantum-assisted variants are optionally evaluated on eligible workloads.
5. Models are validated out-of-time, cross-regime, adversarially and in simulation/paper trading.
6. Accepted models are signed, versioned, attested and added to a Model Pack.
7. Reflex-compatible models are exported to bounded runtime formats and distributed to nodes.
8. Live performance, drift and counterfactual performance continuously determine promote/hold/retire decisions.

11. Quantum Foundry

Quantum is a heterogeneous accelerator, not a magical replacement for classical compute. Every quantum workload has a classical baseline, a common objective and a promotion threshold. QPU jobs are asynchronous and never sit on the market hot path. In v11.6 it can also serve as an experimental co-processor for structured symbolic/combinatorial reasoning, scenario selection and world-model ensemble search when those problems admit useful encodings. Quantum output is a candidate solution or search aid, never a proof of truth; evidence and classical verification remain authoritative.

Workload	Quantum role	Output
Portfolio/family allocation	QUBO/QAOA or other hybrid optimizer	Capital allocation under cardinality, diversification and exposure constraints.
Multi-leg path selection	QUBO/graph optimization	Select feasible cycle/path sets under inventory, settlement, capital and concurrency constraints.
Capital placement	Combinatorial optimization	Where collateral/cash should sit across regions/venues given expected opportunities and transfer cost.
Feature/ensemble selection	Quantum-assisted binary selection	Choose compact learner/feature sets for specialist or reflex models.

Workload	Quantum role	Output
Quantum kernels / variational models	QML research	Evaluate whether quantum embeddings or variational circuits add measurable classification/regression value.
Quantum reservoir/extreme learning	QML research	Experimental representation/learning path for selected time-series or nonlinear tasks.
Scenario/hypothesis selection	Optimization/research	Choose diverse high-information scenarios/experiments under limited compute/data budgets.
Evidence graph clustering	Experimental	Potentially accelerate structured clustering/selection problems; never used as a truth oracle.
Symbolic/combinatorial reasoning search	Experimental hybrid QUBO/QAOA/search	Explore candidate constraint solutions, graph selections or symbolic hypothesis subsets when the formulation is suitable; compare to classical solvers.
World-model ensemble / scenario selection	Experimental optimization/sampling	Select diverse, high-value model/scenario subsets under compute budgets while preserving uncertainty and minority hypotheses.
Hypothesis-space exploration	Research only	Test whether quantum-assisted sampling/optimization can improve exploration of structured hypothesis spaces or experiment portfolios; never treated as empirical validation.
Intervention portfolio / sequence search	Combinatorial optimization / research	Search large permitted action sets, timing choices and resource allocations under constraints. Candidate plans are re-scored by classical causal models, policy gates and simulation before any execution.
Counterfactual branch selection	Scenario/ensemble search	Select high-information alternate futures and intervention/no-intervention branches for expensive simulation or experiment design; no quantum result is treated as causal proof by itself.

Reality boundary: Quantum machine learning and quantum advantage remain experimental for many practical ML workloads. Production use requires benchmarked improvement versus classical methods on the same objective, cost and latency constraints. If quantum loses, the classical result ships.

12. Asset Management Brain

The Asset Brain owns the economic lifecycle of positions, not execution mechanics. It maintains mandates, target exposures, valuation confidence, investment horizon, expected liquidity, hedges, corporate actions, commitments, exit plans and attribution across all supported classes.

- Portfolio construction across strategies, asset classes, regions and horizons.
- Lifecycle: discover -> evaluate -> acquire -> hold/hedge -> monitor -> resize -> exit/divest -> settle -> attribute.
- Illiquid/private mode with commitments, calls, marks, lockups and secondary-exit logic.
- Physical mode integrated with inventory/logistics and product-level economics.
- Continuous reconciliation between economic exposure and legal/settled ownership.

13. Capital Brain & Capital Bank

The Capital Brain decides where deployable capital should be, what it should be doing, and how much liquidity must remain uncommitted. The Capital Bank is the treasury/funding fabric that holds, routes, converts, collateralizes and finances capital. Deposit-taking, lending to third parties, brokerage, derivatives dealing and similar regulated functions are enabled only through appropriate legal entities, counterparties and jurisdictional permissions.

- Cash, stable-value assets, short-duration instruments and idle-cash yield routing.
- Liquidity ladder and withdrawal/reserve forecasts.
- Cross-margin, collateral and encumbrance model.
- FX funding, borrow/lend/repo/credit-line adapters where permitted.
- Capital grants to strategies/regions with expiry, capacity and drawdown limits.
- Cross-region pre-positioning for arbitrage and market making.
- Autonomous treasury transfer inside signed corridors and hard caps.

14. Risk Brain + Deterministic Risk Gate

The Risk Brain forecasts and reasons; the Risk Gate enforces. This split is essential. The brain can use complex AI/causal/scenario models to discover hidden concentration and emerging risk, but every executable action still passes deterministic limits using locally available state.

Risk domain	Examples
Market	Price/volatility/gap/basis/greeks/factor and causal-driver exposure.
Liquidity	Depth, time-to-exit, impact, crowded exit, quote/borrow availability.
Counterparty/Venue	Credit, custody, exchange, bridge, broker and settlement exposure.

Risk domain	Examples
Capital	Leverage, margin, collateral, concentration, funding and liquidity buffers.
Model	Drift, calibration, regime mismatch, uncertainty, feature failure and adversarial inputs.
Data	Stale/poisoned/manipulated/missing/inconsistent source risk.
Operational	Node, network, clock, deployment, credential, reconciliation and dependency failure.
Systemic	Cross-asset/region/counterparty contagion and correlated causal mechanisms.
Conduct / Intervention	Truthfulness, disclosure, identity, communications permissions, market-conduct, manipulation risk, causal overclaiming, unintended externalities and action-channel compromise.

15. Ledger, Accounting & State of Truth

The ledger is an append-only, event-sourced, double-entry multi-asset accounting system. It is authoritative for cash, positions, inventory, obligations, fees, tax lots, collateral, margin, commitments, transfers and settlement state. Market data can disappear and models can be retrained; the ledger cannot be reconstructed from guesses.

- Global journal with idempotent event IDs and immutable lineage from intent -> order -> fill -> settlement -> position/cash impact.
- Subledgers for trading, treasury, custody, physical inventory/commerce, fees/taxes, private commitments and user/mandate ownership.
- Continuous external reconciliation with venues, banks, custodians, wallets, warehouses/marketplaces and settlement agents.
- Every model/strategy/capital/risk decision stores its version identifiers for complete explainability and replay.

16. Regional Reflex Nodes

A Reflex Node is the fast nervous system for one execution region. It does not wait for the global brain. It runs a signed package of models, strategies, beliefs, feasibility rules, capital grants, risk limits, target inventory and opportunity definitions, and combines that package with live local state.

Each Reflex Node embeds a non-blocking Rust fabric client and local append-only event journal. The decision thread never waits on the fabric: it writes to preallocated lock-free queues/journals and continues. Durable outcomes, telemetry and policy/model acknowledgments are drained asynchronously to the regional fabric. Cross-region arbitrage coordination that is truly latency-sensitive remains direct peer-to-peer Reflex Mesh traffic rather than broker-mediated messaging.

Module	Responsibility
Feed adapters	Native venue feeds, normalization, sequence/gap detection, timestamping.
Local books/state	Order books, trades, funding, borrow, venue health, inventory and reservations.
Feature engine	Incremental bounded-cost features.
Reflex inference	Tiny local models and calibrated scores.
Strategy runtime	Compiled strategy predicates and opportunity detectors.
Arbitrage scanner	Whitelisted and discovered graph cycles; local and distributed opportunities.
Intent netting	Cross-strategy netting and internal crossing.
Risk/feasibility gate	Local hard constraints, capacity, inventory, limits and expected edge after costs.
Router/execution	Order types, venue selection, leg sequencing, quote engine and cancel/replace.
Peer mesh client	Fast state summaries, opportunity tokens, reservations and saga messages.
Event journal	Durable local event stream forwarded to ledger/learning asynchronously.

17. Global Reflex Mesh & 3-20-Leg Arbitrage

The Reflex Mesh and the Event & Control Fabric are deliberately separate. Reflex Mesh messages are short-lived, latency-sensitive peer state used to coordinate live multi-leg opportunities. Fabric messages are durable/replayable control, journal and outcome records. A fabric outage must not stall a live local decision, and a Reflex Mesh outage must not prevent durable journaling or control distribution.

The earlier "no cross-region call" rule is too absolute for the product vision. The corrected rule is: no remote dependency is required for a local hot-path decision, but reflex nodes may exchange latency-sensitive state and coordinate distributed opportunities. Cross-region physics cannot be removed, so the architecture uses pre-positioned capital/inventory, synchronized opportunity epochs, bounded TTLs and saga-style recovery rather than pretending distant legs are atomic.

17.1 Arbitrage Graph

Represent every tradable conversion as a directed edge. A node is (asset, venue, representation, settlement stage, region). Edges include price/rate, fees, size-dependent slippage, carry/funding, borrow cost, transfer/bridge/logistics cost, tax/duty estimate, settlement risk, probability of fill and time. Cycles of 2-20 legs are evaluated for executable expected value, not theoretical spread.

17.2 Cross-Region Execution Protocol

1. Global/Warm Arbitrage Brain continuously discovers candidate cycle templates and prepositions inventory/capital.

2. Reflex nodes publish compressed executable edge states - not entire raw books - to peers or regional coordinators.
3. An opportunity epoch identifies the cycle, legs, size, minimum edge, model versions, TTL and unwind policy.
4. Each participating node performs local feasibility and reserves required inventory/capital atomically inside its region.
5. Coordinator or deterministic distributed rule declares FIRE only when the configured reservation condition is met.
6. Legs execute concurrently or in risk-ordered sequence according to cycle policy.
7. Fills are streamed peer-to-peer; partial fills trigger resize, hedge, reroute or unwind state transitions.
8. Completion is recorded to the global ledger; inventory drift becomes a treasury/capital rebalancing task.
9. Every failed/declined/partial cycle is counterfactually scored so the system learns whether the coordination rule was too conservative.

Physics boundary: A 20-leg cycle spanning distant venues cannot be made truly atomic. The platform must price coordination latency, leg risk and unwind cost into the opportunity. When required guarantees cannot be achieved, the cycle is rejected rather than treated as riskless arbitrage.

18. Execution, Market Making & Market Creation

Execution is a venue mesh rather than a broker-specific layer. Venue adapters expose a common capability contract but retain native features. The platform supports order taking, market making, liquidity provision, cross-venue routing, derivatives execution, event contracts, decentralized venues and physical marketplaces where legally and operationally supported.

18.1 Market Making

- Dynamic fair value, spread, skew and size from volatility, inventory, queue value, adverse-selection and confidence.
- Portfolio-aware quoting so hedges and correlated exposure affect quote size before a fill occurs.
- Quote-rate/message-ratio controls, cancel/replace budgets and venue-specific microstructure models.
- Capital and inventory shared with arbitrage through one reservation fabric.

18.2 Market Creation / Market Factory

Where a venue and jurisdiction allow it, Algorik may create/list a market or instrument, seed initial liquidity, publish pricing rules, manage inventory and continuously measure whether the market remains economically and legally valid. Market creation is a governed product workflow, not an unrestricted model action.

19. Prediction, Event & Regulated Wagering Markets

Prediction/event markets are first-class because they convert world beliefs into directly tradable probability. The platform can ingest event definitions, resolve source-of-truth rules, estimate outcome distributions, price contracts, trade, market-make and - where permitted - create new markets. Sports or other wagering venues are treated as a separate regulated venue type with jurisdiction, age/eligibility and product controls; the core cognition can price events without assuming every venue is legally available to every entity/user.

20. Physical Commerce & Product Arbitrage

"Physical" must be more than a valuation class. A system that can purchase and resell products needs a commerce operating layer. It tracks SKU identity, condition, location, inventory, shipping, insurance, customs/duties, tax, marketplace fees, payment/chargeback risk, storage and returns.

Domain	Responsibility
Product Scout	Find price/availability differences, auctions, liquidations, wholesale listings and supply shocks.
SKU/Entity Resolver	Map equivalent products across retailers/marketplaces/regions and detect variants/counterfeits.
Commerce Feasibility	All-in landed cost, expected resale, time-to-sale, return risk, inventory carry and capital lockup.
Purchase Executor	Merchant/marketplace adapters with limits, account controls and fraud safeguards.
Logistics Brain	Route, carrier, consolidation, warehouse, customs, insurance and delivery risk.
Inventory Ledger	Physical units, condition, location, cost basis, reservations and ownership.
Resale Router	Choose marketplace/auction/direct venue; price and reprice inventory.

21. Asset & Market Coverage

Class	Support model
Cash / FX	Fiat, spot/forward FX, cash equivalents, funding and currency conversion.
Equities / ETFs / indices	Listed equities, ETFs, baskets, indices and corporate actions.
Rates / fixed income / credit	Government/corporate debt, yield curves, credit, repo/funding interfaces where available.
Commodities	Energy, metals, agriculture, futures, spot/physical links, shipping/storage economics.
Derivatives	Options, futures, perpetuals and other permitted derivatives with margin/greeks/settlement models.
Crypto / tokenized	Spot, derivatives, tokenized representations, DEX/CEX venues, bridges and on-chain settlement.
Prediction/event	Event contracts and probability markets.

Class	Support model
Private/illiquid	Private equity/credit/funds, real assets and other non-continuously priced investments.
Physical products	SKU-level commerce, auctions, collectibles, wholesale/retail arbitrage and inventory.
Created markets/instruments	Only where supported by venue/product/legal governance.

22. Continuous Learning & Evaluation Brain

Every path becomes training data, including paths not taken. The Evaluation Brain is independent of the Strategy/Model Foundry so that the creator of an idea is not the sole judge of the idea.

- Live outcome attribution by strategy, model, belief, evidence source, capital policy, risk rule and execution path.
- Counterfactual replay for declined, partially filled, hedged and unwound opportunities.
- Champion/challenger models and strategies with regime-conditioned scorecards.
- Data-source alpha attribution: did this source improve decisions after cost and delay?
- Quantum-vs-classical benchmarking recorded per workload family and instance size.
- Automatic retirement of degraded models/strategies and rollback to last known-good artifact.
- Closed-loop research prompts generated from failure clusters and unexplained residuals.
- Novelty and knowledge-gap detection: identify unexplained residuals, low-confidence regions, contradictory evidence, missing causal links, unsupported asset classes and capabilities with no competent specialist.
- Regional reflex contribution: microstructure drift, local failures, missed fills, queue behavior and venue anomalies become globally shareable episodes rather than staying trapped in one region.
- Memory consolidation and compression: preserve high-value episodes and abstractions while deduplicating stale detail so learning can compound without memory growth becoming unbounded noise.
- Tick-learning loop: live fills, slippage, queue errors, missed/expired opportunities and book/flow regime changes are reconciled against replay assumptions and become microstructure retraining/evaluation data.
- World-market reaction loop: every material world event is paired with observed multi-asset/tick response over multiple horizons; these reaction episodes improve causal propagation models and teach the system when markets ignore, anticipate or overreact to information.

23. Intelligence Expansion Engine - Open-Ended Learning

The system must be able to become more capable without requiring a new architecture every time it encounters a new market, data domain, model family or tool. "Open-ended" does not mean uncontrolled self-modification. It means the platform is architected so its knowledge, ontology, memories, specialists, tools, models and research agenda can expand continuously, while every new capability is evaluated, sandboxed, lineage-tracked and promoted through explicit gates.

The expansion engine sits above the Scout, Evidence, Memory, World Model, Model Foundry, Quantum Foundry, Evaluation Brain and Reflex Mesh. It converts ignorance, surprise and failure into structured research work, then converts proven research into deployable capability.

Expansion mechanism	How intelligence grows	Primary artifact
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World-coverage expansion	Scout agents continuously discover new countries, venues, APIs, registries, datasets, sensors, products, chains, languages and lawful information surfaces.	New SourceSpec / FeedPack
Evidence-quality expansion	Source reputation, provenance graphs, corroboration, contradiction resolution and adversarial detection improve as outcomes reveal which information channels were reliable.	Updated EvidencePolicy / SourceScore
Memory expansion + consolidation	Semantic, episodic, procedural, market and failure memories grow from experience, then consolidate, summarize and deduplicate so signal compounds without preserving every raw detail forever.	MemoryPack / EpisodeCluster
World-model + ontology expansion	Novel entities, relationships, causal drivers, asset classes, event types and physical processes extend the knowledge graph and ontology when existing schemas cannot explain observations.	OntologyPatch / GraphSchema
Specialist-brain creation	When a domain repeatedly produces unresolved questions, the platform can instantiate a new specialist agent/model with its own tools, corpus, evaluations and authority scope.	BrainSpec / SpecialistPack
Cross-domain synthesis	A synthesis layer searches for causal and economic links across domains - e.g., weather -> shipping -> commodity basis -> FX -> equities -> options -> event markets.	CrossDomainHypothesis
Continuous training + active learning	Outcomes, drift, uncertainty and information value select what data to label, simulate, collect or retrain on next instead of retraining blindly.	TrainingCampaign
Simulation + counterfactual expansion	Digital twins, market replay, scenario generation, agent simulations and counterfactual worlds create experience for rare regimes and paths the live system did not take.	ScenarioPack / SyntheticEpisode
Model evolution + distillation	Champion/challenger models, ensembles, new architectures and larger slow-lane models are continuously tested; useful capability is distilled into smaller warm/reflex artifacts.	ModelPack lineage
Autonomous research agents	Research agents form hypotheses, gather evidence, write experiments, build features, run backtests, inspect failures and propose new strategies or models.	ResearchCampaign / ExperimentPack
Quantum-assisted research	Quantum resources can be used for benchmarked optimization, graph search, feature/ensemble selection, kernels/variational experiments and other hybrid learning research where they add measurable value.	QuantumExperiment / Benchmark
Tool and capability acquisition	The system maintains a Tool Registry and can propose connectors, parsers, solvers, simulators, code modules, data adapters and external model interfaces required to answer questions it currently cannot answer.	ToolSpec / CapabilityPack

Reflex-node local learning	Each reflex cell records local microstructure, venue behavior, latency, fill/slippage, anomaly and failure episodes; useful lessons propagate to global training while latency-critical inference stays local.	RegionalEpisode / ReflexDelta
Collective/federated intelligence	Generalist cognition, specialist brains, evaluation agents and regional nodes share typed knowledge artifacts rather than isolated chat/state, allowing one component's discovery to improve the whole system.	KnowledgeDelta / SharedEpisode
Self-evaluation + calibration	The platform measures forecast accuracy, uncertainty calibration, P&L attribution, execution quality, model decay, source value, hallucination/error rates and unexplained residuals.	Scorecard / GapSignal
Self-generated curriculum	A Curriculum Engine ranks what the system should learn next based on uncertainty, economic value, novelty, failure frequency, strategic importance and expected information gain.	CurriculumItem / ResearchQueue
Market-microstructure expansion	Continuously learn new venue mechanics, queue behavior, liquidity regimes, cross-asset lead/lag, execution dynamics and market reactions from tick/order-book history plus live Algorik telemetry; generate replay/retraining tasks when residuals grow.	TickCampaign / MicrostructureDelta
World-model population expansion	When one representation becomes insufficient, the platform can spawn additional domain, regional, horizon, scenario or counterfactual world models and benchmark their incremental predictive/explanatory value.	WorldModelSpec / WorldModelBranch
Symbolic theory growth	Validated patterns can become new ontology relations, rules, invariants, causal structures, constraints or proof procedures; contradictions generate explicit theory-revision tasks.	RulePack / ConstraintPack / TheoryDelta
Ambient proactive expansion	Always-on models discover new weak signals, precursor relationships, assumption failures and high-value unknowns continuously; useful ambient signals can wake specialists or create curriculum items without a human prompt.	AmbientSignal / AttentionEvent

23.1 Recursive Intelligence Expansion Loop

1. Observe novelty: detect surprise, low confidence, contradictions, missed opportunity, unexplained P&L, model drift or a request that no current brain/tool can satisfy.
2. Classify the gap: data gap, ontology gap, causal gap, memory gap, model gap, tool gap, specialist gap, execution gap or compute/quantum research opportunity.

3. Generate curriculum: estimate economic value, risk, novelty, information gain, compute/data cost and urgency; rank the research task.
4. Acquire capability: discover lawful data, create a connector/tool, instantiate a specialist, select models/solvers, or construct simulations needed to investigate the gap.
5. Research and experiment: run reproducible campaigns across historical, simulated, paper and shadow environments with provenance and lineage.
6. Evaluate independently: the Evaluation Brain tests calibration, robustness, leakage, regime sensitivity, cost, latency, safety, legal eligibility and incremental value versus current champions.
7. Promote by lane: successful capability may remain a slow-lane research tool, become a warm service, or be distilled/compiled into a bounded reflex artifact. Promotion is never direct from research into live money movement.
8. Learn from production: outcomes, failures, vetoes, missed opportunities and local reflex episodes update memory, source scores, world-model beliefs and future curriculum.
9. Repeat: every resolved gap exposes new gaps, creating a compounding research and capability loop.

23.2 Registries That Prevent Intelligence From Becoming Fixed

- Source Registry - every known feed, endpoint, document family, sensor, marketplace and discovery method with provenance, cost, license, freshness and utility.
- Ontology Registry - versioned entity, event, relationship, market, asset, product, causal-driver and lifecycle types; supports governed schema extension.
- Memory Registry - semantic knowledge, episodes, procedures, failures, simulations and compressed abstractions with retention/value policy.
- Brain Registry - generalist and specialist agents/models, their competency domains, tools, evaluations, calibration, authority and version lineage.
- Tool Registry - connectors, browsers, parsers, solvers, code runtimes, simulators, geospatial tools, venue adapters, model endpoints and quantum backends.
- Model Registry - foundation, specialist, ensemble, policy and reflex models plus datasets, features, calibration, benchmarks, resource budgets and rollback ancestry.
- Research Registry - hypotheses, experiments, null results, open questions, benchmark suites and unresolved contradictions so the system does not repeatedly rediscover the same failures.
- Capability Registry - what the platform can currently sense, reason about, simulate, trade, settle, hedge, transfer, purchase, create or operate - and the confidence/eligibility boundary for each capability.

23.3 Self-Generated Curriculum and Curiosity

The Curriculum Engine gives the platform a deliberate mechanism for learning what it does not know. It should not optimize for novelty alone. Each candidate research task is scored on expected economic value, reduction in uncertainty, strategic coverage, risk reduction, information gain, reuse across domains, compute/data cost and legal/operational feasibility.

- Unknown unknown surfacing: clusters of anomalies, residuals and contradictory beliefs create investigation tasks even when no human asked a question.
- Coverage pressure: asset classes, geographies, languages, venues or causal drivers with weak evidence/model coverage automatically generate acquisition and research work.
- Failure-driven curriculum: repeated vetoes, adverse selection, failed arbitrage sagas, reconciliation breaks and incorrect forecasts become explicit learning objectives.

- Opportunity-driven curriculum: persistent price/belief discrepancies can trigger creation of a specialist, simulator, new data feed or model family to determine whether the discrepancy is real.
- Capability-gap curriculum: when a task requires a missing parser, API, simulator, model interface or quantum/classical solver, the missing tool itself becomes a build/research task.
- Market-learning curriculum: unexplained slippage, fill/queue prediction errors, new venue mechanics, abnormal lead/lag or world-event reaction residuals automatically create targeted tick-replay and microstructure research campaigns.

23.4 Controlled Self-Improvement and Promotion

The platform may propose and build new code, tools, agents, features, schemas and models, but it does not mutate the production control plane or money-moving path in place. Every change moves through an isolated branch/sandbox, reproducible tests, security and data-policy checks, simulation/replay, independent evaluation, shadow/paper operation and signed promotion. The more authority a capability receives, the stronger its evidence and rollback requirements become.

- Research freedom is broad; execution authority is narrow and explicit.
- New tools default to read-only/sandboxed permissions until their behavior and data handling are understood.
- New specialists have declared competency boundaries and abstain outside them.
- A new model must beat a named baseline on a declared benchmark and remain calibrated under stress/regime shift.
- A new reflex artifact must meet deterministic resource, latency and failure-mode budgets before regional deployment.
- Every promoted capability keeps lineage to the research campaign, datasets, code, model, evaluations and rollback parent that produced it.

23.5 What "Open-Ended" Means

The design is intentionally open-ended: its intelligence is not capped by a predefined list of markets, sources, agents or model families. It can keep expanding as new data, compute, algorithms, tools and domains become available. This is not literal infinite compute or guaranteed intelligence growth. Expansion remains bounded by physics, capital, compute, data rights, regulation, security, benchmark quality and the requirement that new capability demonstrate measurable value before receiving authority.

24. Causal Agency & Intervention Intelligence

Prediction is incomplete agency. A cognitive financial system should also understand which variables it can legitimately influence, which actions are available to it, how those actions propagate through the world, and whether the observed outcome was actually caused by the intervention rather than merely correlated with it. The Causal Agency & Intervention Engine therefore sits between cognition and tool execution. It turns desired outcomes into testable, policy-constrained causal action plans and turns action results back into learned causal structure.

24.1 Desired Outcomes, Utility and Constraints

A desired outcome is represented as a typed GoalSpec rather than a free-form instruction. It declares the target state/distribution, entities affected, time horizon, success metric, acceptable uncertainty, budget, risk envelope, legal/jurisdictional boundary, identity that may act, prohibited side effects and stop/rollback conditions. Goals

can be financial, operational, research, product, liquidity, risk-reduction, information-acquisition or communications objectives.

The system separates outcome preference from action permission. Wanting an outcome never implies permission to use every causal lever that might achieve it.

24.2 Action-Affordance & Causal Lever Graph

The World Model Federation is extended with an action-affordance graph: which variables Algorik can observe, which it can control, which tools can affect them, expected latency/cost, causal paths to outcomes, known side effects, dependencies and authority requirements. Symbolic reasoning enforces hard constraints while learned causal models estimate uncertain effect sizes.

Interventions are represented explicitly using structural causal models and counterfactual branches where practical: do(action) -> intermediate states -> target outcome. When causal identification is weak, the engine abstains, gathers more evidence or proposes an experiment rather than pretending correlation is control.

24.3 Intervention Planner

The planner generates multiple candidate action sequences, including a no-action baseline. It simulates each across competing world models, market/tick digital twins and scenario branches, then compares expected causal effect, confidence, cost, capital usage, time-to-effect, reversibility, legal eligibility, conduct risk and downside.

Candidate interventions can be parallel, sequential, conditional or adaptive. A plan may say: gather evidence first; if condition A is confirmed, publish an attributable research note through an authorized channel; if market/liquidity state B then hedge or provide liquidity; if response C occurs, stop or switch strategy.

24.4 Authorized Action Surfaces

Algorik acts only through registered tools with explicit identities, permissions and audit. Example action surfaces include:

- Truthful publishing and communications - attributable articles, research notes, disclosures, reports or posts through authorized channels with source lineage, factuality checks, disclosure policy and immutable audit.
- Authorized stakeholder outreach - messages, requests for information, customer/vendor/partner engagement or negotiation through approved identities and channels.
- Research actions - acquire lawful data, run surveys/experiments where permitted, request additional evidence, launch simulations or commission a specialist research campaign.
- Product and pricing actions - adjust Algorik-controlled product configuration, pricing, availability or service behavior where governance permits.
- Operational actions - routing, logistics, inventory, compute, deployment, scheduling, sourcing and other owned-system changes.
- Capital and risk actions - allocate, hedge, rebalance, reserve liquidity, move collateral or reduce exposure within signed authority envelopes.
- Market/product actions - quote, provide liquidity, create/list/seed permitted markets or instruments, or execute commerce/trading actions subject to venue, jurisdiction and market-conduct controls.

24.5 Communications Integrity & Anti-Manipulation Boundary

External communication is an intervention surface, not an exemption from evidence and conduct rules. The Communications & Conduct Brain checks factual support, provenance, identity, required disclosures, audience/channel permissions, jurisdiction and whether the proposed pattern could constitute deceptive coordination or market manipulation.

The platform must not use fabricated identities, sockpuppets, bots posing as independent people, fake consensus/astroturfing, false or misleading claims, undisclosed paid promotion, coordinated rumor manufacture, wash trading, spoofing, pump-and-dump behavior or other deceptive techniques to move prices or beliefs. If the causal plan depends on deception, the plan is infeasible.

24.6 Experimentation and Causal Effect Attribution

After an intervention, the Evaluation Brain estimates what changed because of the action. Where feasible this uses randomized or staged experiments; elsewhere it uses matched controls, synthetic controls, difference-in-differences, instrumental/structural methods, Bayesian causal models, counterfactual world-model branches and explicit no-action baselines. The platform records uncertainty and never upgrades correlation into causal certainty without evidence.

Effect is measured at multiple horizons. A communication may change knowledge immediately, behavior later and market/operational outcomes still later. The attribution object preserves the causal chain, confounders, competing explanations and confidence rather than recording only a final KPI.

24.7 Ambient Agency Loop

Ambient models can proactively surface a DesiredOutcome candidate or ActionOpportunity when they detect a risk, opportunity, information gap or controllable causal lever. That does not directly cause action. The Attention Router sends it to the Agency Brain, which builds a GoalSpec and intervention candidates, runs simulation and policy/conduct gates, and obtains whatever authority the action class requires.

The recursive loop is: Observe -> predict -> specify desired state -> identify controllable levers -> generate interventions -> simulate against many worlds -> verify authority/conduct -> act -> observe -> attribute causal effect -> update world models, action policies and future intervention priors.

24.8 What the System Learns About Agency

The platform maintains an Action Outcome Memory containing intervention context, audience/environment state, action sequence, causal hypothesis, predicted effect, actual observations, attribution confidence, side effects and no-action counterfactual. Over time it learns which actions work, for whom, under which regimes, with what delay, and when not acting is the superior intervention.

25. Governance, Legal Isolation & Authority

A full-capability system must be able to connect to many regulated domains without assuming one legal entity is allowed to do all of them. Technical capability and legal permission are separate dimensions. Every venue, product, user/entity, jurisdiction and strategy carries eligibility metadata. The same cognitive core can evaluate an opportunity that the execution layer is prohibited from acting on. The same separation applies to causal agency: a cognitive model may analyze an intervention that the acting identity, entity, channel or jurisdiction is not permitted to perform.

- Jurisdiction/product eligibility gate before strategy deployment and before every new venue/instrument becomes executable.
- Separate identities and credentials for trading, custody/transfer, market creation, banking/financing and commerce purchasing.
- Human-signed or governance-signed authority envelopes for destinations, leverage, product classes and regulated functions.
- Immutable audit of model, data, policy, code and control versions involved in each decision.
- Research/simulation may be broader than executable scope; unsupported opportunities remain shadow-only.

- Every action tool has a registered acting identity, permission scope, disclosure policy, audience/channel rules, rate/budget limits and revocation path.
- Goal approval, plan approval and action execution are separately auditable; sensitive intervention classes can require stronger approval than ordinary research.
- Public communications and market-facing actions are cross-checked for market-conduct/manipulation risk and cannot be justified solely by expected profit or desired price impact.

26. Platform & Deployment Architecture

The target deployment preserves the existing cloud direction: Google Cloud for application/data/compute and IBM Quantum for QPU workloads. The primary architectural correction is workload placement by lane rather than trying to use one runtime everywhere.

Workload	Target placement
Regional Reflex Cells	Dedicated GCE C3/C3D-class low-latency VMs or equivalent; Rust; CPU pinning/huge pages/mlock/busy polling as justified; direct venue connectivity.
Warm services	GKE for stateful/long-running platform services; Cloud Run where stateless burst/serverless economics are beneficial.
Training/AGI	GPU/TPU compute on Google Cloud; Python/JAX/PyTorch or other fit-for-purpose tooling; artifact export to bounded inference runtimes.
Global ledger	Cloud Spanner or equivalent strongly consistent multi-region transactional store plus append-only object/event archive for immutable evidence.
Native event/control fabric	Algorik Rust Event & Control Fabric on dedicated regional GCE broker/controller nodes across three zones; QUIC/mTLS native protocol, replicated partition logs, selective cross-region mirrors, GCS archival and typed Rust SDK. No Google Pub/Sub or Kafka on the internal core path; optional compatibility bridges are edge adapters only.
Object/data campaigns	Cloud Storage for temporary/retained artifacts, manifests, model packages, simulation outputs and reproducible campaign datasets.
Secrets/keys	Cloud KMS/HSM plus segregated custody/signing systems; no browser access to keys, ledger internals or execution nodes.
Quantum	IBM Quantum via isolated Quantum Gateway/Foundry with queue-aware asynchronous job orchestration and classical fallback.
Observability	OpenTelemetry + OpenObserve/compatible stack; trace each end-to-end opportunity across brain, model, risk, node, venue, ledger and learning loop.

Workload	Target placement
Tick archive + replay	Cloud Storage/object archive for tick/internal execution history with compressed partitioning, manifests and entitlements; replay workers use dedicated compute and local high-throughput caches/indexes. External world source content is re-fetched from its origin/provider when needed and is never retained as a raw corpus.
Microstructure training	GPU/CPU/accelerator training pools on Google Cloud; distributed feature/replay jobs; outputs packaged through the Model Registry into warm/reflex-compatible artifacts.
World Model Federation	Long-running versioned world-model services on GKE/compute pools with domain-specific stores/graphs; no single model is on the reflex critical path.
Symbolic reasoning services	Graph/logic/constraint/planning runtimes in warm/slow compute pools; deterministic solvers may also compile bounded predicates into reflex artifacts.
Ambient model mesh	Continuously scheduled/streaming model services with event-driven wakeups, stateful forecasts, surprise scoring and attention routing; isolated resource budgets prevent background cognition from starving trading paths.
Causal Agency / Action Orchestrator	Slow/warm services on GKE/Cloud Run with typed tool adapters, policy/identity service, causal simulation access and durable action journal. Public/channel adapters use segregated credentials and disclosure/conduct policy; no direct path from ambient model to external action.
Effect Attribution / Experiment Service	Batch/stream causal-evaluation workers joined to episode/world-model stores; supports experiment assignment where permissible, synthetic/no-action controls, multi-horizon metrics and uncertainty-aware EffectAttribution outputs.

26.1 Native Rust Event & Control Fabric

Algorik uses a native Rust messaging and journal subsystem as its internal asynchronous nervous system. It replaces Google Pub/Sub and core Kafka dependencies so control semantics, replay, partitioning, backpressure, regional independence and binary/runtime behavior can be designed around Algorik instead of around a generic broker. The fabric is intentionally not the execution hot path: local market-event-to-order decisions and direct peer Reflex Mesh coordination remain independent.

Component	Responsibility
fabricd brokers	Regional partition leaders/followers; append-only logs, subscriptions, consumer checkpoints, retention, replication and backpressure.

fabric-controller	Three-zone metadata quorum for stream definitions, partition placement, broker epochs, ACL metadata and leadership. Metadata uses a Rust Raft implementation; no cloud database is required for continued regional operation.
fabric-mirror	Selective asynchronous cross-region replication. No global write quorum; each region remains independently writable/operational.
fabric-schema	Versioned Protobuf descriptors/contracts, compatibility checks, schema hashes and generated Rust types.
fabric-archive	Sealed segment export to Cloud Storage/Tick Lake with manifests, hashes, entitlements and replay indexes.
fabric-bridge	Optional edge-only adapters for external Kafka/Pub/Sub/webhooks when integration requires them. Internal services use the native protocol.
fabricctl / fabric-inspect	Rust operator CLI for stream/partition health, lag, ACLs, mirroring, replay and emergency isolation.
fabric-bench / fabric-chaos	Latency/throughput benchmark harness and deterministic fault/partition testing used in CI and game days.
Rust SDK	Zero-copy-friendly producer/consumer APIs, idempotent producer sequencing, consumer groups, async subscriptions, local ring-buffer integration and OpenTelemetry hooks.

Protocol and semantics

- Transport: QUIC over UDP with mutual TLS for broker/client and cross-region mirror traffic; long-lived streams for durable records and QUIC datagrams only for explicitly lossy telemetry classes.
- Wire format: versioned Protobuf via prost for contracts and compatibility; BLAKE3/content hashes and CRC protection on records/segments. Hot in-process data structures may use Rust-native zero-copy layouts but never leak unversioned memory formats onto the network.
- Ordering: strict per-partition order. No false promise of a total global order. Every record carries source event time, receive time, region, partition, leader epoch, offset and hybrid logical clock metadata.
- Delivery: at-most-once allowed only for telemetry; at-least-once is default for durable streams; idempotent producer IDs/sequence numbers and deterministic event IDs provide effectively-once processing with idempotent sinks.
- Durability classes: Control uses three-zone quorum plus persistent metadata/WAL; financial outcomes use quorum replication and archival; market journals use high-throughput replicated segments; telemetry can be sampled/dropped under pressure.
- Backpressure: every stream has explicit byte/message quotas, lag limits and overload policy. Critical control/outcomes cannot be silently dropped; telemetry and low-value ambient signals can be sampled or shed.
- Cross-region behavior: selected streams mirror asynchronously over Google private networking. A regional cluster never blocks on remote acknowledgement, preventing a remote outage from stopping local execution or control consumption.

Hot-path contract

The Reflex Node publishes through a non-blocking local ring/journal writer running on a separate core. Order-book processing, model inference, risk veto and venue send never wait on a broker acknowledgement. Model and policy packages are stored as immutable signed artifacts in object/artifact storage; the fabric carries only

version/digest/availability/activation messages. If the fabric is unavailable, a node continues under its last valid package and local journal until the package TTL or risk policy requires degradation.

27. Core Contracts Between Brains

Contract	Required fields/purpose
EvidencePacket	Claim, entities, time/location, source/provenance, license, confidence, contradiction links, content hash.
BeliefState	Proposition/distribution, evidence set, causal path, confidence, uncertainty type, TTL.
ModelPack	Signed model artifacts, features, calibration, allowed universes, resource budget, expiry, rollback parent.
StrategyPack	Compiled strategies, dependencies, hypotheses, universes, capacity, performance envelope, retirement rules.
CapitalGrant	Owner/mandate, region/strategy, amount, collateral class, leverage, expiry, drawdown and liquidity constraints.
RiskEnvelope	Hard limits, scenario limits, causal-driver limits, model/data health requirements, emergency mode.
OpportunityToken	Cycle/market ID, legs, size, edge, TTL, reservations, execution policy, unwind/hedge policy, version lineage.
LedgerEvent	Immutable economic event with debit/credit impacts, ownership, settlement state and full attribution chain.
Episode	World state, beliefs, actions/declines, outcome horizons, surprise and counterfactuals.
MarketEvent	Canonical event with venue/instrument/type, source sequence, exchange/event time, receive time, normalized time + uncertainty, payload lineage/hash and entitlement.
MarketStatePack	Time-bounded reconstructed book/flow/liquidity/regime state or learned embedding with integrity flags, feature lineage and horizon.
ReplayManifest	Dataset partitions, venue/instrument universe, time window, clock/latency/queue/fill/impact assumptions, model versions, random seeds and reproducibility hashes.
MicrostructureDelta	Observed live-vs-model residuals for fills, slippage, impact, liquidity, lead/lag or venue behavior that can trigger evaluation/retraining/curriculum work.
WorldModelState	Model identity/scope/horizon, entities/state distributions, causal structure, evidence lineage, assumptions, confidence/calibration, branch parent, update time and expiry.
ModelDisagreement	Competing model/hypothesis IDs, disputed propositions, supporting evidence, calibration/history, unresolved assumptions and arbitration status.
SymbolicDerivation	Goal/conclusion, rules/constraints/graph paths used, evidence references, solver/proof result, violated assumptions, uncertainty and machine-checkable trace.

Contract	Required fields/purpose
AmbientSignal	Signal class, observed deviation, expected baseline, horizon, novelty/surprise, affected entities/assets, urgency, suggested wake targets, evidence lineage and expiry.
AttentionEvent	Reason for specialist/reflex/research activation, originating ambient/world/symbolic models, budget, priority, authority boundary and completion/feedback link.
GoalSpec	Desired state/distribution, target entities, horizon, utility/success metric, budget, risk/side-effect constraints, identity, jurisdiction, authority class, prohibited methods and stop/rollback conditions.
InterventionPlan	Causal hypothesis, controllable levers, ordered/conditional candidate actions, tool identities, expected effect distribution, cost, timing, reversibility, world-model simulations, no-action baseline, required approvals and abort rules.
ActionIntent	One authorized external/internal action with acting identity, channel/tool, payload/parameters, factual/provenance references if communicative, disclosure requirements, policy decisions, expiry/idempotency and audit lineage.
EffectAttribution	Goal/intervention/action IDs, observations across horizons, control/counterfactual method, estimated causal effect distribution, confounders, side effects, confidence/identifiability and world-model/action-policy updates.
GapSignal	Observed surprise/uncertainty/failure, affected domains, evidence, severity, economic/risk value and candidate gap class.
CurriculumItem	Research question, expected information/economic value, required tools/data, budget, owner agent/brain, evaluation suite and stop conditions.
BrainSpec	Competency domain, tools, source permissions, memory scope, model stack, evaluations, abstention rules and authority envelope.
ToolSpec	Capability, interface, permission scope, data handling, dependency/security provenance, sandbox tests, cost/latency and allowed callers.
KnowledgeDelta	Versioned ontology/world-model/memory update with provenance, confidence, conflicts, originating brain/region and merge policy.
FabricEnvelope	Stream namespace, schema/version hash, region, partition, producer ID/sequence, leader epoch, offset, source/receive/HLC timestamps, priority/QoS, trace ID, idempotency key, payload hash and auth context.

Contract	Required fields/purpose
StreamPolicy	Partition key, replication factor, ack policy, retention, durability/QoS class, byte/message limits, consumer lag threshold, mirroring policy, overload/shedding behavior and entitlement.
ControlPackAnnouncement	Artifact kind, immutable URI/digest, version, signer/attestation, target regions/nodes, activation window, expiry, rollback parent and required acknowledgments.
MirrorCheckpoint	Source/destination region, stream/partition, source epoch/offset, destination offset, lag, integrity hash and last successful mirror time.

28. Failure and Degradation Model

The system should degrade by capability, not fail as one monolith.

Failure	Required behavior
Global cognition unavailable	Regional nodes continue with last valid packs until expiry; size narrows as confidence/TTL decay. No new research/policy.
Peer reflex mesh degraded	Local trading continues; distributed cycles requiring coordination are disabled or reduced to pre-positioned directional bands.
Quantum unavailable	Classical solver/model path runs; no execution interruption.
Data source poisoned/stale	Evidence quarantined; dependent beliefs/models downweighted or invalidated.
Ledger degraded	New money-risking actions halt if authoritative accounting cannot be committed; safe cancels/flattening continue under local journal rules.
Custody/treasury unavailable	Trading constrained to available settled capital; no new transfer dependence.
Venue failure	Quarantine venue, cancel exposure if possible, reroute/hedge elsewhere, reconcile asynchronously.
Model drift	Fall back to challenger/baseline model or deterministic strategy; shrink capital automatically.
Tick-feed sequence gap / corrupt book	Mark affected interval/instrument unreliable, request snapshot/recovery, quarantine derived features if needed, and prevent contaminated windows from training/replay until integrity is restored.
Clock synchronization degraded	Preserve raw timestamps, widen temporal uncertainty, disable strategies/research requiring precise cross-venue ordering, and block precision-sensitive labels/replay from promotion.
Replay model diverges from live execution	Reduce confidence/capital on dependent strategies, recalibrate digital twin with live telemetry, rerun benchmark suite and require renewed promotion before restoring authority.

Failure	Required behavior
World models diverge	Do not force false consensus. Preserve branches, raise uncertainty, reduce authority where appropriate, request discriminating evidence/simulations and let arbitration use calibration + outcome history.
Symbolic contradiction / unsatisfied constraints	Mark conclusion/action infeasible or unresolved, retain the contradiction trace, search alternate hypotheses/plans and fail closed for money-moving invariants.
Ambient model storm / noisy proactive alerts	Rate-limit and budget attention, cluster duplicates, penalize low-value signal producers, quarantine drifting ambient models and protect reflex/ledger resources from background cognition.
Causal effect not identifiable	Do not claim the action caused the outcome. Downgrade confidence, preserve competing explanations, seek discriminating evidence or design a safer experiment; do not scale autonomy from correlation alone.
Action/channel identity or permission compromised	Immediately revoke/quarantine the adapter and credentials, stop pending actions, preserve audit, rotate identity/secrets and require explicit re-authorization. Never substitute an unapproved identity/channel.
Communication/conduct gate fails	Block the external action, retain the failed plan and reasons, and search for a truthful/non-manipulative alternative or abstain. Profitability never overrides conduct constraints.
Intervention causes adverse side effect	Trigger plan-specific stop/rollback/hedge/mitigation actions where feasible, reduce the relevant action policy authority, create a failure episode and require re-evaluation before reuse.
Regional Event Fabric broker loss	Partition leadership moves to an in-sync replica in another zone. Producers retry by epoch; consumers resume from committed offsets. Reflex execution does not wait for failover.
Regional Event Fabric quorum lost	Local Reflex Nodes continue from cached signed policy and local journals; new control changes stop; durable event publication queues locally within bounded disk budgets; policy-defined degradation occurs before journal exhaustion.
Cross-region fabric mirror down	Local streams remain writable. Mirror lag is visible and bounded; no cross-region synchronous quorum is attempted. Global consumers mark remote knowledge stale and replay from regional/GCS archive when connectivity returns.

29. Autonomous Platform Operations / AIOps

- AI observes infrastructure, applications, data quality, model health and trading behavior as one graph.
- Every deploy runs simulation, integration, security, performance and paper-trading gates before promotion.
- Autonomous remediation may restart, rollback, scale, fail over, quarantine or disable a capability inside pre-approved runbooks.
- Incident agent correlates telemetry, deployment changes, data anomalies, venue behavior and model drift into one timeline.
- SLOs are defined for market-data freshness, decision latency, order/fill integrity, ledger commit, reconciliation, model serving, data provenance and quantum job success.
- Action observability traces GoalSpec -> InterventionPlan -> approvals -> ActionIntent -> external/internal effects -> EffectAttribution as one lineage graph.
- Communications/channel health includes identity integrity, delivery/reach metrics, provenance/disclosure checks, policy violations and anomalous coordination patterns.
- Causal-policy drift is monitored: when observed intervention effects deviate from predicted uplift or side-effect assumptions, autonomy shrinks automatically.

30. Build Order

Trying to build every capability at once would produce an impressive diagram and an untestable system. The architecture should be complete from day one, but implementation should be gated.

Phase	Exit capability
Phase 0 - Foundations	Canonical types/contracts, ledger, simulation harness, observability, CI/CD/GitOps, identity/security, venue adapter SDK. Native Rust Event & Control Fabric contracts, 3-zone regional quorum, Rust SDK and benchmark harness are part of foundations.
Phase 1 - One Reflex Cell	One asset/venue, local fast path, deterministic risk, ledger, paper trading, replay. Canonical tick capture, book reconstruction and deterministic replay for that first market are mandatory.
Phase 2 - Cognitive Data Loop	Scout Fabric, evidence/provenance, world model, episodic memory, Market Intelligence & Tick Learning Fabric, Tick Lake, replay/digital twin, world+market fusion, model/strategy foundry. Establish initial World Model Federation, model-disagreement representation and symbolic/temporal/causal reasoning services.
Phase 3 - Multi-Asset + Capital/Risk	Asset registry, Asset/Capital/Risk brains, treasury, custody corridors, multiple asset classes.
Phase 4 - Multi-Region Reflex Mesh	Americas/Europe/APAC nodes, peer mesh, opportunity tokens, distributed reservations, 2-5 leg cycles. Add cross-region tick alignment, lead/lag learning and distributed opportunity replay. Add selective cross-region fabric mirroring for outcomes/control while keeping latency-sensitive arbitrage state on direct Reflex Mesh links.

Phase	Exit capability
Phase 5 - 3-20 Leg Arbitrage + Market Making	Graph optimization, pre-positioned inventory, saga recovery, quote engine, cross-class hedge map.
Phase 6 - Quantum Foundry	Classical benchmark first, then quantum optimization and QML research plug-ins; ship only measured wins. Add benchmarked experiments for symbolic/combinatorial search and world-model/scenario ensemble selection.
Phase 7 - Prediction/Market Factory/Commerce	Event markets, market creation workflows, physical product commerce and logistics adapters.
Phase 8 - Financial AGI Autonomy	Broader tool use, self-directed research campaigns, specialist-agent society, autonomous model/strategy lifecycle under authority envelopes. Add the Ambient Model Mesh and Attention Router so cognition proactively predicts, scans and launches research without prompts.
Phase 9 - Open-Ended Intelligence Expansion	Curriculum Engine, ontology/source/tool/brain/capability registries, sandboxed tool/agent creation, active learning, memory consolidation, regional-to-global learning and continuous benchmark-gated promotion. Include market-microstructure curricula driven by live-vs-replay residuals and newly discovered venues/market structures. Enable governed spawning/retirement of new world models, symbolic theory/rule growth and ambient-signal-driven curricula.
Phase 10 - Causal Agency & Intervention Intelligence	GoalSpec/action-affordance graph, Intervention Planner, authorized action adapters, communications/conduct gate, causal effect attribution, experiment/no-action baselines, Action Outcome Memory and ambient-to-agency routing. Begin shadow-only, then allow narrow reversible actions before broader authority.

31. Capability Completeness Checklist

- **COMPLETE TARGET:** Fast reflex lane and slow cognitive lane are explicit and independently deployable.
- **COMPLETE TARGET:** Warm coordination lane connects slow and fast systems without putting global services on the local hot path.
- **COMPLETE TARGET:** Global data discovery can autonomously find, assess, register, feed and retire sources.
- **COMPLETE TARGET:** World information is evidence-scored and provenance-tracked; no component treats scraping as truth by default.
- **COMPLETE TARGET:** World model, causal graph, episodic/semantic/procedural memory, counterfactuals and self-model exist.
- **COMPLETE TARGET:** Generalist/AGI-style planners and specialist agents can run in the slow lane and use tools under policy.

- **COMPLETE TARGET:** Model Foundry supports large models, specialist models and distilled reflex models.
- **COMPLETE TARGET:** Quantum Foundry supports optimization and experimental learning, benchmarked against classical baselines.
- **COMPLETE TARGET:** Asset, Capital, Risk, Treasury, Ledger and Settlement brains are separate first-class domains.
- **COMPLETE TARGET:** Regional reflex nodes can operate independently and communicate with peers.
- **COMPLETE TARGET:** 2-20-leg arbitrage is modeled as an executable graph with reservations, coordination, hedging and unwind semantics.
- **COMPLETE TARGET:** Market making and governed market creation are supported.
- **COMPLETE TARGET:** Prediction/event markets are first-class; regulated wagering is isolated by eligibility/jurisdiction.
- **COMPLETE TARGET:** Physical product purchase/resale has commerce, logistics and inventory capabilities, not just a price model.
- **COMPLETE TARGET:** All major liquid, event-driven, private/illiquid, tokenized and physical classes have an extensible registry.
- **COMPLETE TARGET:** Ledger is authoritative and double-entry across cash, positions, obligations, collateral and physical inventory.
- **COMPLETE TARGET:** Continuous evaluation learns from actions, declines, partial fills and failures.
- **COMPLETE TARGET:** AIOps can operate and remediate the platform autonomously within runbooks and safety envelopes.
- **COMPLETE TARGET:** Legal capability gates are separate from technical capability so the same brain can research more broadly than it can execute.
- **COMPLETE TARGET:** Every dependency has a defined degraded mode; quantum and global cognition can never block the reflex path.
- **COMPLETE TARGET:** An Intelligence Expansion Engine turns novelty, uncertainty, failure and unsupported requests into ranked research/capability work.
- **COMPLETE TARGET:** Source, ontology, memory, brain, tool, model, research and capability registries are versioned and extensible rather than hard-coded.
- **COMPLETE TARGET:** The platform can create new specialist brains and tool integrations in sandboxes, independently evaluate them and promote only measured improvements.
- **COMPLETE TARGET:** Self-generated curriculum/active learning determines what data, simulations, labels, experiments and research campaigns should run next.
- **COMPLETE TARGET:** Regional reflex experience is merged into global learning through typed episodes/knowledge deltas without adding a synchronous hot-path dependency.
- **COMPLETE TARGET:** Memory consolidation, ontology expansion and cross-domain synthesis let intelligence compound instead of only accumulating raw data.
- **COMPLETE TARGET:** Quantum-assisted learning/search remains an expandable research path but never bypasses classical benchmarks, evidence rules or production promotion gates.
- **COMPLETE TARGET:** Raw and canonical tick/order-book history is a governed training asset with reproducible manifests, entitlements and deterministic reconstruction.
- **COMPLETE TARGET:** Historical replay models latency, queues, fees, rejects, partial fills, impact and multi-leg failure/unwind rather than assuming fills from bars.
- **COMPLETE TARGET:** World intelligence and market microstructure intelligence are time-aligned and fused; event causality can be compared with actual market propagation.
- **COMPLETE TARGET:** Microstructure/execution models are continuously recalibrated from live Algorik telemetry and distilled into bounded reflex artifacts only after independent evaluation.

- COMPLETE TARGET: World intelligence is a federation of numerous domain, regional, horizon, scenario and counterfactual world models rather than one monolithic model; disagreement and branch lineage are explicit.
- COMPLETE TARGET: A Symbolic & Neuro-Symbolic Reasoning Fabric provides ontology, temporal, causal, rule, probabilistic, constraint, planning and proof-trace reasoning alongside neural models.
- COMPLETE TARGET: Proactive ambient models run continuously, maintain forecasts and surprise scores, monitor assumptions, search weak signals/opportunities and wake specialist/research pathways without prompts.
- COMPLETE TARGET: Quantum research can assist selected structured symbolic/combinatorial search and world-model/scenario ensemble selection, but every result is benchmarked and independently verified.
- COMPLETE TARGET: Ambient cognition has explicit attention/resource budgets and cannot starve the reflex, risk, ledger or custody paths.
- COMPLETE TARGET: Desired outcomes are explicit GoalSpec objects with success metrics, horizons, budgets, side-effect constraints, acting identity and prohibited methods rather than free-form wishes.
- COMPLETE TARGET: A Causal Agency & Intervention Engine maps goals to controllable causal levers, generates multiple action/no-action plans, simulates them against competing world models and executes only through typed authorized tools.
- COMPLETE TARGET: External communication is an audited action surface with truthfulness, provenance, identity, disclosure, channel/jurisdiction and market-conduct checks; deceptive/manipulative influence methods are infeasible by policy.
- COMPLETE TARGET: Intervention learning estimates causal effect and uncertainty across multiple horizons using experiments or defensible observational methods; correlation alone cannot grant more autonomy.
- COMPLETE TARGET: Action Outcome Memory and EffectAttribution update world models, causal graphs, ambient priors and action policies so the system learns which interventions work, where, when and with what side effects.
- COMPLETE TARGET: Ambient models may propose action opportunities but cannot directly perform external actions; the Agency Brain, authority and conduct gates sit between detection and execution.
- COMPLETE TARGET: Quantum-assisted intervention search may rank large constrained action/scenario sets but cannot bypass classical causal validation, policy, identity, conduct or authorization gates.
- COMPLETE TARGET: Google Pub/Sub and core Kafka are not internal runtime dependencies; the Algorik Rust Event & Control Fabric provides typed publish/subscribe, replicated journals, replay, backpressure and selective mirrors.
- COMPLETE TARGET: A fabric outage cannot block market-event-to-order execution; Reflex Nodes use non-blocking local journal/ring writers and cached signed control packages.
- COMPLETE TARGET: Direct Reflex Mesh traffic remains separate from durable event fabric traffic, so multi-leg coordination can optimize latency independently of replay/durability concerns.
- COMPLETE TARGET: Every durable event stream declares partitioning, ordering, retention, replication, overload and mirroring semantics rather than relying on unspecified broker defaults.

32. Technology Reality Notes (2026)

The design intentionally distinguishes product ambition from current hardware maturity. IBM documents QAOA as a hybrid quantum-classical optimization workflow and provides utility-scale examples. IBM Research continues active quantum-machine-learning work, while published surveys still identify practical limitations including data encoding, circuit/noise behavior and gradient/training challenges. Therefore the Quantum Foundry is architecturally broad but operationally benchmark-gated. The multi-world, neuro-symbolic and ambient-model architecture is a system-design target rather than a claim that any single current model already

provides general financial intelligence. Its purpose is to let heterogeneous reasoners and model populations compound while preserving explicit evaluation, uncertainty and authority boundaries. Causal intervention is similarly benchmark-gated: reliable agency requires identifiable causal structure and measured effects, not merely high predictive accuracy. Quantum search may help explore candidate action/scenario spaces, but current quantum methods do not establish causal truth or authorize real-world interventions.

- IBM Quantum Documentation - Quantum Approximate Optimization Algorithm (QAOA), quantum.cloud.ibm.com.
- IBM Research - Quantum Machine Learning research program, research.ibm.com/topics/quantum-machine-learning.
- IBM Research / Physical Review Applied - survey of near-term QML techniques and limitations (2024).
- IBM Quantum Documentation - hybrid quantum-enhanced ensemble classification workflow and Qiskit Functions.

33. Final Definition of Algorik

Algorik is a globally distributed cognitive financial AGI target system. It continuously discovers and evaluates world information; captures market reality tick by tick; maintains a federation of competing world and causal models; reasons neurally, symbolically and counterfactually; runs proactive ambient models; and uses classical and quantum computation as heterogeneous research and optimization tools. It manages assets, capital, liquidity, collateral and risk; creates and trades opportunities across financial, event and physical markets; coordinates regional reflex nodes for local and multi-leg execution; and maintains one authoritative ledger. Beyond predicting futures, Algorik also models agency: it can represent a desired outcome, identify lawful controllable levers, generate and simulate interventions, act through authorized tools, measure whether the action actually changed the outcome, and learn a better causal action policy. Its intelligence may become increasingly general, but every action that can move money, alter an external system or communicate publicly is bounded by explicit identity, authority, truthfulness, conduct, feasibility, audit and rollback controls.

Its internal asynchronous nervous system is the Algorik Rust Event & Control Fabric: regionally independent, replayable, backpressure-aware and selectively mirrored. The fabric carries control, journals and outcomes, while the local Reflex hot path and direct Reflex Mesh remain independent for deterministic speed and failure isolation.

Architecture mantra: Sense globally. Capture market reality tick by tick. Verify evidence. Maintain many world models. Reason neurally and symbolically. Predict continuously. Specify desired outcomes. Discover controllable causal levers. Simulate interventions. Act only through authorized truthful channels and tools. Measure causal effect. Allocate intelligently. React locally. Coordinate globally. Account exactly. Learn from every actual and counterfactual outcome. Expand capability safely.